ADVANTEST®

V93000 Series

DC Scale DPS128 Cookbook

SmarTest 7.2.3.3
SmarTest 7.3.2.1

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1 DC Scale DPS128

The DC Scale DPS128 is the next generation high density channel slot device power supply focusing on high multisite and multi power-domain applications. It is part of the Smart Scale series and addresses the needs of pattern synchronized, highly efficient and massive parallel resources optimized for DC source and measurement.

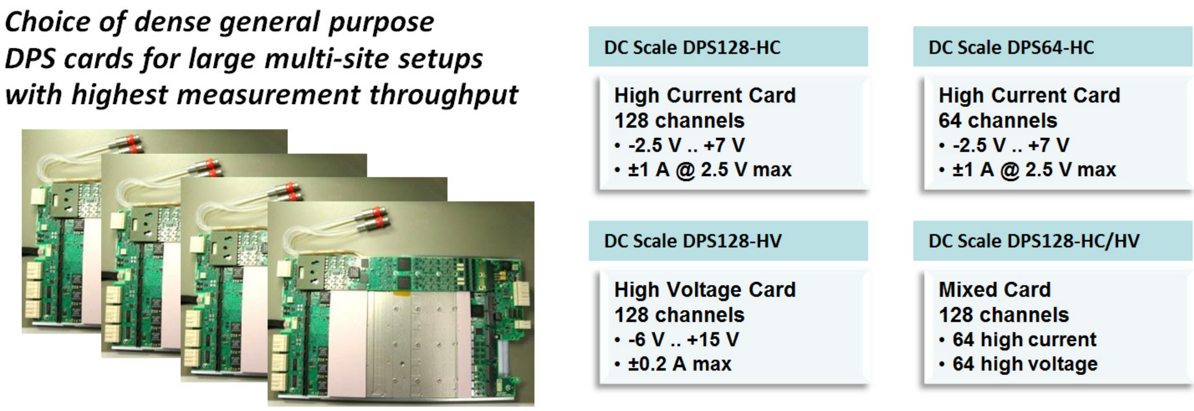


Figure 1: DC Scale DPS128 Cards Overview

As part of the Smart Scale series DC Scale DPS128 works together with all Smart Scale based digital cards as well as all DC Scale cards and the MSDPS power supply.

Basic feature set:

- Implements high current “HC” and high voltage “HV” channel versions.
- DPS and PMU like instrument modes.
- Can force and measure voltage and current.
- Negative voltage support.
- Voltage force ±3 mV accuracy, voltage measure accuracy ±1 mV.
- Multiple Current force and measure ranges from ±1 A down to ±12.5 µA.
- All channels of same type can be ganged, per board, up to 64 ganging groups.
- Un-ganged current measurement mode.
- High impedance voltage measurement mode.
- AWG and digitizing functionality (current and voltage).
- Comparator functionality for monitoring, match-loop support.
- Slew rate control.
- Profiling.
- Cap-less operation support.
- DC Scale compatible working model.

Please refer to TDC topic “DC Scale DPS128/DPS64 Cards” (138572).

1.1 Hardware Versions

DC Scale DPS128 implements 2 channel type versions (“High Current”, “High Voltage”) with focus on different application areas.
In total there are 4 DPS128 board versions available using different combinations of channel types.

2x DPS128 boards with pure “High Current” type of channels:

- **E8023CS:** DPS128-HC - 128 channel HC version
- **E8023CSH:** DPS64-HC - 64 channel HC version

	E8023CSH	E8023CS
Maximum number of channels per board	64	128
Maximum source/sink current per channel	1.0 A @ 2.5 V	1.0 A @ 2.5 V
Maximum source/sink current per channel	0.5 A @ 7 V	0.5 A @ 7 V
Maximum source/sink current per board	64 A / 64 A @ 2.5 V	66 A / 58 A @ 2.5 V
Voltage range	-2.5 V to +7 V	-2.5 V to +7 V

Figure 2: E8023CSH / E8023CS configuration overview

1x DPS128 cards with pure “High Voltage” type of channels:

- **E8023CSW:** DPS128-HV - 128 channel HV version

E8023CSW	High voltage
Maximum number of channels per board	128
Maximum source/sink current per channel	0.2 A @ -6 to +15 V
Maximum source/sink current per board	13.2 A / 24.4 A
Voltage range	-6 V to +15 V

Figure 3: E8023CSW configuration overview

1x DPS128 cards with mixed “High Current”/”High Voltage” type of channels:

- **E8023CSV:** DPS128-HC/HV - 128 channel mixed HC/HV version (64 channels each)

E8023CSV	High current	High voltage
Maximum number of channels per board	64	64
Maximum source/sink current per channel	1.0 A @ 2.5 V	
Maximum source/sink current per channel	0.5 A @ 7 V	0.2 A @ -6 to +15 V
Maximum source/sink current per board	62 A / 54 A @ 2.5 (only high current channels used)	12.8 A / 12.8 A (only high voltage channels used)
Voltage range	-2.5 V to +7 V	-6 V to +15 V

Figure 4: E8023CSV configuration overview

1.2 Pogo Cable Assemblies

DPS128 will support a variety of pogo cable assemblies to ensure DUT board compatibility.

DPS128 “native” type pogo (1 A per channel):

“Native” type pogo (1A per channel):	
E8023PA Supported by: - E8023CS - E8023CSH - E8023CSW - E8023CSV	- 128 channels on pogo side, 1 A @ 2.5 V - 8 IO pogo blocks, 16 channels per block, DPS32 pogo compatible

Using E8023PA together with half populated E8023CSH (DPS64), the first 4 IO pogo blocks will be powered by HC type of channels. The remaining ones will be unused.

Using E8023PA together with mixed E8023CSV (DPS128 HC/HV) the first 4 IO pogo blocks will be powered by HC type of channels. The last 4 IO pogo blocks will be powered by HV type of channels.

The final pogo locations can be controlled by the pogo mapping section in the model file. For further reference see chapter “A-1 Model File” for more details.

DPS128 “compatible” type pogo (>1 A per channel):

Compatible mode pogo in terms of DUT board compatibility and current capability per channel, but with test program incompatibilities.

“Compatible” type pogo (2 A .. 4 A per channel)	
E8023PC Supported by: - E8023CS	- 64 channels on pogo side, 2 A @ 2.5 V (cable ganged) - 4 IO pogo blocks, 16 channels per block, DPS32 pogo compatible
E8023PD Supported by: - E8023CSH	- 32 channels on pogo side, 2 A @ 2.5 V (cable ganged) - 2 IO pogo blocks, 16 channels per block, DPS32 pogo compatible
E8023PE Supported by: - E8023CS	- 64 channels on pogo side, 2 A @ 2.5 V (cable ganged) - 8 MSDPS pogo blocks, 8 channels per block, MSDPS pogo compatible - MSDPS redirect , GULP style
E8023PF Supported by: - E8023CSH	- 16 channels on pogo side, 4 A @ 2.5 V (cable ganged) - 2 MSDPS pogo blocks, 8 channels per block, MSDPS pogo compatible - MSDPS redirect , QUL style
E8023PG Supported by: - E8023CSH	- 32 channels on pogo side, 2 A @ 2.5 V (cable ganged) - 4 MSDPS pogo blocks, 8 channels per block, MSDPS pogo compatible - MSDPS redirect , GULP style

Refer to chapter “A-2 Pogo Cable Details” for more details on pogo layout and channel mapping.

2 DPS128 Feature Set

2.1 DPS128 Basic Feature Listing

DC Scale DPS128 is the next generation general purpose channel slot device power supply. Its HC type of channels will accompany DC Scale DPS32 addressing large multisite and multi power domain applications.
Its HV type of channels will accompany DC Scale VI32 in the micro controller unit "MCU" space as well as automotive and mobile space including power management ICs.

DC Scale DPS128 operates in all 4 quadrants and in doing so, both types of channels support positive and negative voltages as well as current source and sink capabilities.

	MSDPS	DC Scale VI32	DC Scale DPS32	DC Scale DPS64-HC DPS128-HC	DC Scale DPS128-HC/HV DPS128-HV
Ganging	All channels, 16 A	All channels 48 A	All channels, 48 A	All channels, 64 A (DPS64) 66 A (DPS128)	All channels of same type 62 A / 12.8 A (HC/HV) 13.2 A (HV)
Sink Capability	16 A	48 A	48 A	64 A (DPS64) 58 A (DPS128)	54 A / 12.8 A (HC/HV) 24.4 A (HV)
4 Quadrant DPS	Yes	Yes	No	Yes	Yes
High Voltage Support	No	Yes (25 V)	No	No	Yes (15 V, HV channel type)
Pattern Controlled	No	Yes	Yes	Yes	Yes
Slew-Rate Control	Yes	Yes	Yes	Yes	Yes
Voltmeter (high impedance mode)	No	Yes	Yes	Yes	Yes
Differential Voltmeter (measured between sense lines)	No	Yes	Yes	No	No
Cap-less Mode	Not specified	Yes	No	Yes	Yes
Profiling	No	Yes	Yes	Yes	Yes
AWG / Digitizer	limited	Yes	Yes	Yes	Yes
Compare Mode (match-loop)	No	No	No	Yes	Yes (HC channel type) No (HV channel type)
Surge Detector (latch excessive voltage / current)	No	No	No	Yes	Yes (HC channel type) No (HV channel type)

Figure 5: DPS cards feature listing

2.2 DC Scale Instrument Operation Modes

DPS128 implements two instrument operation modes as defined for the DC Scale family of cards:

- DPS instrument mode (DPS)
- PMU instrument mode (SIG)

As common for DC Scale, both modes are supporting “code-based” static programming, as well as “pattern-based” dynamic programming (force and measure events anchored in the vector stream).

2.2.1 DPS Instrument Mode

In DPS instrument mode you can use DC Scale DPS128 cards like a standard DPS, to supply devices with power during the execution of a test flow. Each channel can source and sink current at a programmed voltage or via clamp circuitry, and each channel can measure the voltages and currents supplied.

The DPS channels will take part in standard connect and disconnect sequences and set activation.

Interactive “Hardware Monitor” views as implemented in SmarTest.

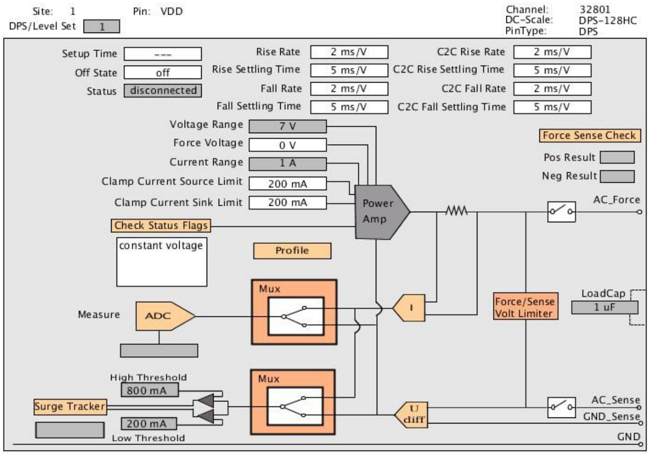


Figure 6: HW Monitor - DPS instrument mode (HC type)

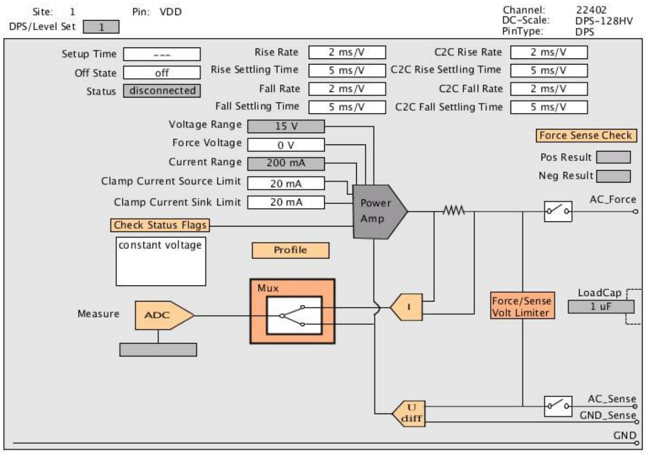


Figure 7: HW Monitor - DPS instrument mode (HV type)

2.2.2 PMU Instrument Mode

In PMU instrument mode you can use DC Scale cards like a standard PMU to temporarily supply power in a single testsuite or for a group of testsuites. Each of the 128 channels can source and sink current at a programmed voltage or via clamp circuitry, and each channel can measure the voltages and currents supplied.

Channels defined in the PMU mode do not take part in any standard connect and disconnect sequences, and cannot take part in the set activation, although you can add them to user-defined connect and disconnect sequences.

Interactive “Hardware Monitor” views as implemented in SmarTest.

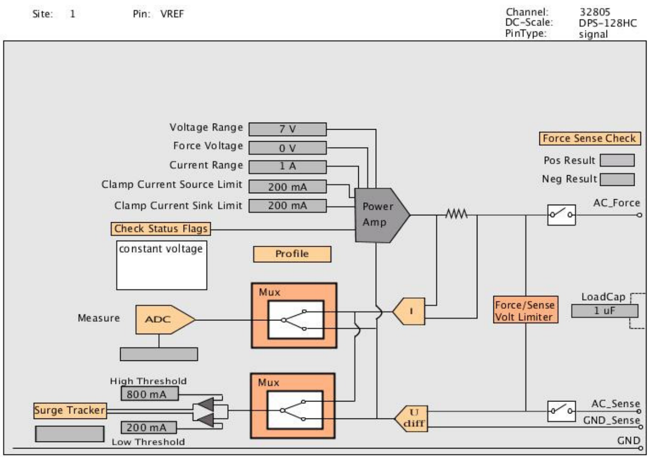


Figure 8: HW Monitor – PMU instrument mode (HC type)

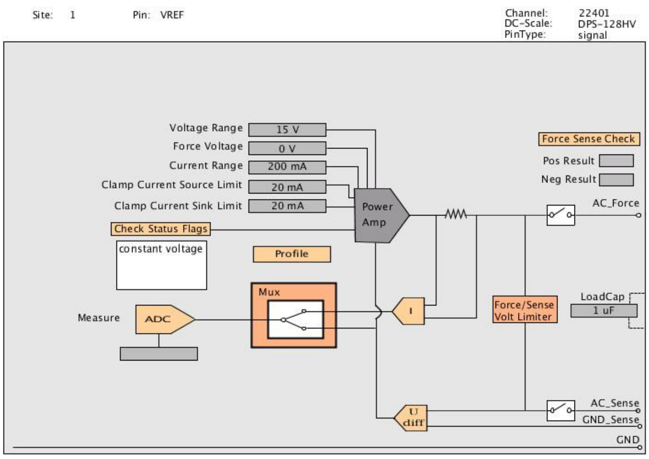


Figure 9: HW Monitor - PMU instrument mode (HV type)

2.3 Sequencer Controlled DC – Dynamic DC

2.3.1 Concept

Dynamic DC is a feature that allows triggering DC events within a pattern. DC events can be a voltage or current force actions, but also measurement actions.

The benefit of Dynamic DC is that these events are vector synchronous to any digital pin operation. Also pattern based operations allow to burst multiple DC events within a sequencer run, while in parallel adapting any digital channel.
Bursting DC events creates a significant throughput benefit, as there is no need for repeating instrument and device conditioning, but also allows for a combined result upload of all events.

The concept of Dynamic DC can be explained by the following graphics.

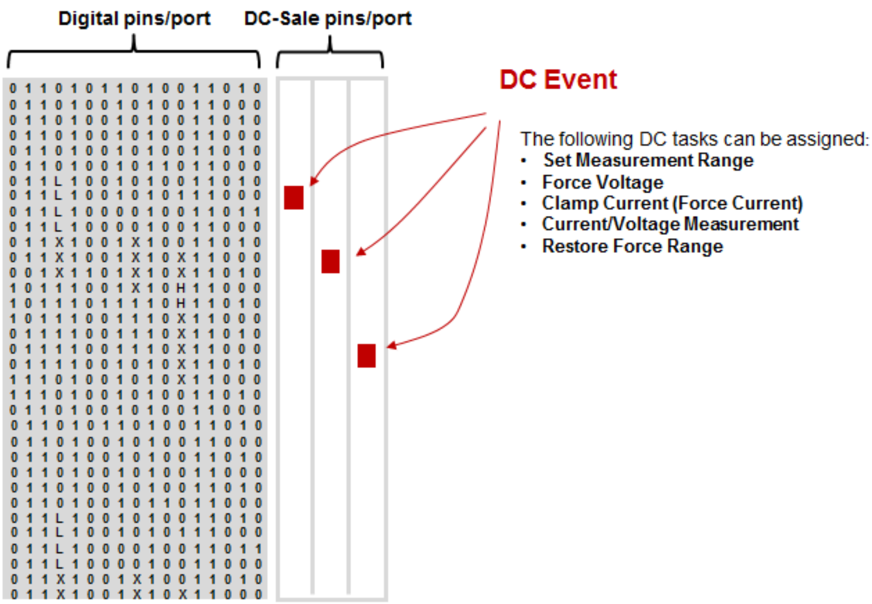


Figure 10 : Dynamic DC events in pattern stream

Digital Channel:
During the pattern execution each vector is referring to a physical waveform that is sent to the DUT. The physical waveform can drive 0 or 1 or compare the logical level of device output.

DC Scale channel:
During the pattern execution dc events can be executed at certain vector position. A DC event can set a measurement range, force voltage/current or perform voltage/current measurements. The minimum anchor distance between consecutive DC events is 32 vectors.

The DC event is referred to as test data, while the vector position is referred to as the test point. So, the concept of Dynamic DC is to define a set of test data as an event which describes what to do at the vector, and to anchor an event as a test point in a pattern.

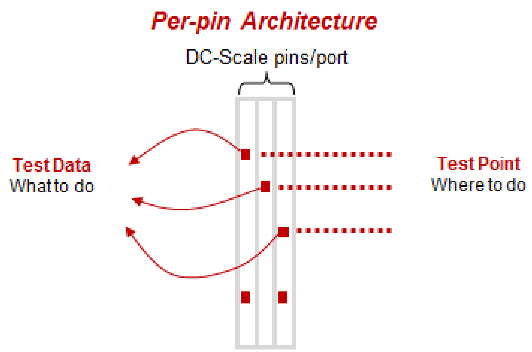


Figure 11: Concept of test data and test point

2.3.2 Sequencer Controlled Measurement Events

DC Scale DPS128 supports dynamic voltage and current measurement events.

To execute any Dynamic DC measurements it is mandatory to first setup the instrument to the correct measurement mode. This is done by adding a range event upfront. There is a dedicated range event for current measurement and voltage measurement.
If multiple measurements are done within the same range it is enough to call the range event just once at the beginning of the sequence.
It is possible to switch between current and voltage measurements within a pattern run. In this case it is required to also switch the range to the corresponding measurement mode.

In the case of the current measurement it is sometimes required to restore the original current range after the measurement within the pattern. A special restore event is provided to do so.

The execution of Dynamic DC measurements is as simple as executing a functional test. When the pass limit has been declared in the event definition, the measurement will be judged instantly at each anchor. The result of each judgment will also affect the overall functional test result. That is to say, if any of the DC measurements fail, the result of the functional test will fail as well. Furthermore, since pass/fail judgment is done in the DC Scale channels without involvement of any data upload there will be no extra overhead introduced.

The value of each measurement can be uploaded by calling the Testmethod API "`DC_RESULT_ACCESSOR`". The API uploads all the data of all pins and sites in one shot for further processing. This scheme reduces the uploading overhead.

For further reference see chapter "Current Ranges & Current Force/Measure Scenarios" for more details regarding current range settings and required settling times.

2.3.3 Sequencer Controlled Force Events

DC Scale DPS128 supports dynamic voltage and current force events.

In contradiction to measure events the force events are restricted to the defined voltage or current force range. It is not possible to excess the range, as changing the force range itself is not supported within a pattern run (requires disconnect sequence).

Force events require a minimum execution time of 2.7µs before any other event can be issued. Additionally each force event has to be followed by 8 vectors to guarantee proper processing by test processor (i.e. any force event should not be placed within an 8 vector distance to a sublabel call or/and not within the last 8 vectors of any pattern). Minimum anchor distance in the pattern is 32 vectors.

In the DPS instrument mode the ranges are defined by the actual level primary set.
In the PMU instrument mode the ranges are typically defined during the PMU connect sequence. This is within Testmethod context utilizing "`SPMU_TASK`" API.

For more information on how to connect in PMU instrument mode refer to chapter "PMU Instrument Mode" or "Testmethod Programming".

For further reference see chapter "Current Force" for more details regarding dynamic current force operations.

2.3.4 Programming Model

Dynamic DC events can be programmed in the Testmethod context or directly via Pattern Editor. The use- / programming model is the same for DPS and PMU instrument mode.

Programming via Pattern Editor persistently saves the event data to the level setup file (test data) and the pattern setup file (test point).

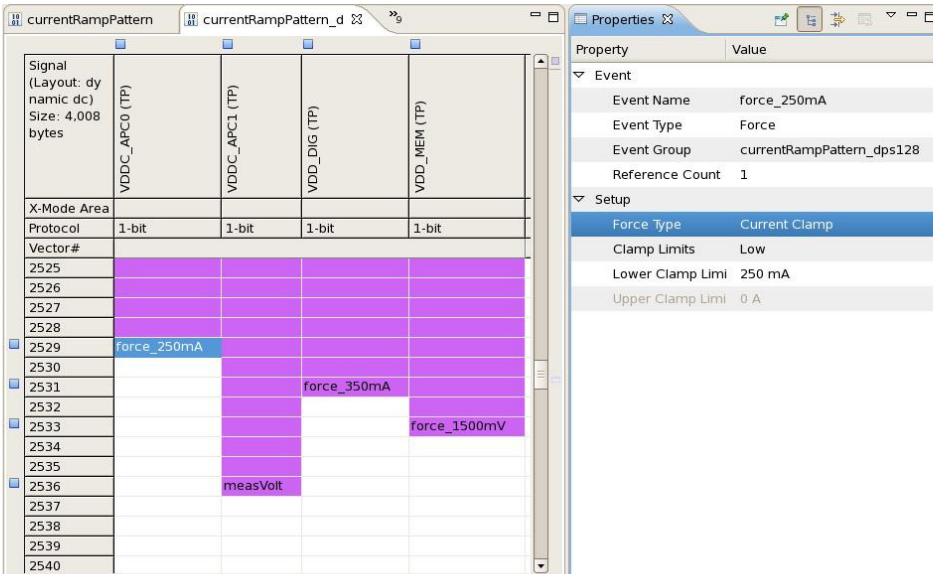


Figure 12: Event setup in Pattern Editor

Programming via Testmethod dynamically creates the setup. This is done by the means of "EVENT" and "PATTERN_CONTROLLER" Testmethod API.

```
// test data
EVENT_VM measVolt("measVolt");
measVolt.limitLow(4.9 V).limitHigh(5.1 V).averages(8);

PATTERN_CONTROLLER controller;
controller.label("dcscale_label","dps128_port")
                   .vecEvents("dps128_pins")
                   .clear()
                   .insert(200,measVolt)  // (test point, test data)
                   .setup();
```

Figure 13: Event setup in Testmethod

Note that the setup algorithm should be optimized in a way to only create the events once (i.e. during first call / testflow run).

"EVENT" API listing (not complete):

Range Events:

- EVENT_RANGE_IM
- EVENT_RANGE_VM
- EVENT_RANGE_RESTORE

Force Events:

- EVENT_IC (→ current force via current clamp)
- EVENT_VF
- EVENT_WAVEFORM_IC
- EVENT_WAVEFORM_VF

Measure Events:

- EVENT_IM
- EVENT_VM
- EVENT_WAVEFORM_IM
- EVENT_WAVEFORM_VM

For a complete list of "EVENT" APIs refer to TDC topic 116911.

Minimum execution times (settling times) for range and measure events are variable and are listed in chapter "Current Measurement Scenario". In all cases a minimum of 8 vectors is required after event placement. Minimum anchor distance in the pattern is 32 vectors.

Programming examples can be found in the Chapter "Testmethod Programming".

2.4 Comparators

Each DPS128 HC type of channel has a pair of comparators which can be connected to the voltage or current sensor. It allows for monitoring the voltage on the sense line or the output current of the channel.

In voltage compare mode the DC accuracy is around ±10 mV (typical) with a minimum detectable pulse width of 1µs.
In current compare mode the DC accuracy is ±10 % (typical) of the selected current range with a minimum detectable pulse width in dependency to the range. It is 1µs for the highest range (1A) and scaling by factor 10 for each 2nd range going from the highest to lower ranges (i.e. 200mA, 25mA, 2.5mA, …).

Comparators are addressed via special operation modes, controlled by new Testmethod API "`DPS_AUX_TASK`".

This feature is not available for the DPS128 HV type of channel(s).

2.4.1 Surge Tracker

Surge Tracker is a monitoring feature to detect current or voltage surges over a specified limit by monitoring the voltage via the sense line as well as the output current of the channel.
Some flags are latching onto the status if any voltage or current surge goes beyond the thresholds. The detection of a surge is bipolar.

- High (positive) and low (negative) thresholds can be set
- Two flags are latched for high and low surges respectively

Once the tracker is turned on, it keeps on monitoring unless it is being turned off explicitly, i.e. some connect / disconnect sequence does not reset it.

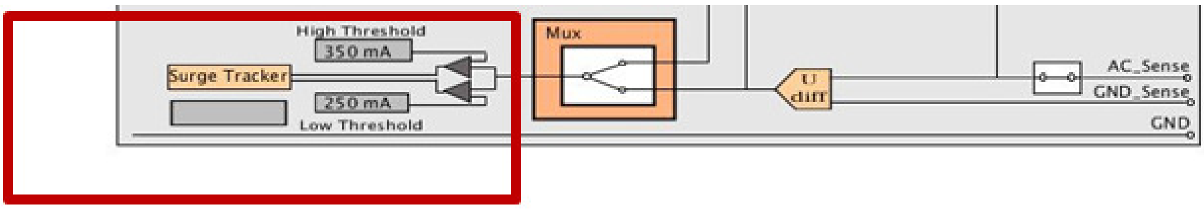


Figure 14: Surge Tracker in Hardware Monitor

Refer to chapter "Testmethod Programming" for more details.

2.4.2 Match-Loops

The comparator provides some basic compare event which enables Match-loop like applications. The implementation follows the same rules as for the existing implementation on digital channels.

- Match-loop is controlled by sequencer instructions like “Match Activate” and “Match Repeat” within the pattern
- Maximum supported pattern frequency is 50Mhz (with minimum detectable pulse width of 1µs or lower)

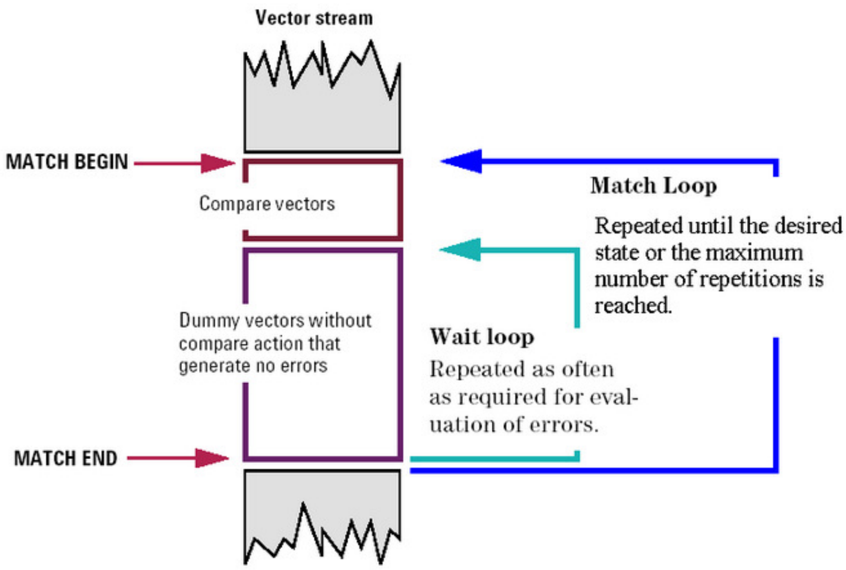


Figure 15: Match-Loop concept: TDC topic 122701

Compare event characteristics:

- Works for DPS and PMU mode
- Works for current and voltage
- Single threshold compare against high threshold
- Compare mode (lower than/higher than) can be changed within the pattern (per event)
- Compare type (voltage/current) can be changed within the pattern (per event)
- Compare type is independent of currently selected measurement range (if any)
- Compare event has a fixed length of 32 vectors (minimum distance requirement for consecutive compares), while the active compare event is located at vector 32

The threshold and compare type setup is programmed via “`DPS_AUX_TASK`” Testmethod API. Compare events (low/high) are either programmed directly within the Pattern Editor or by using “`EVENT_CMP`” Testmethod API.

It is not possible to query the result state of the compare event; therefore the effective use of the event is restricted to match-loop like applications.

Refer to chapter “Testmethod Programming” for more details.

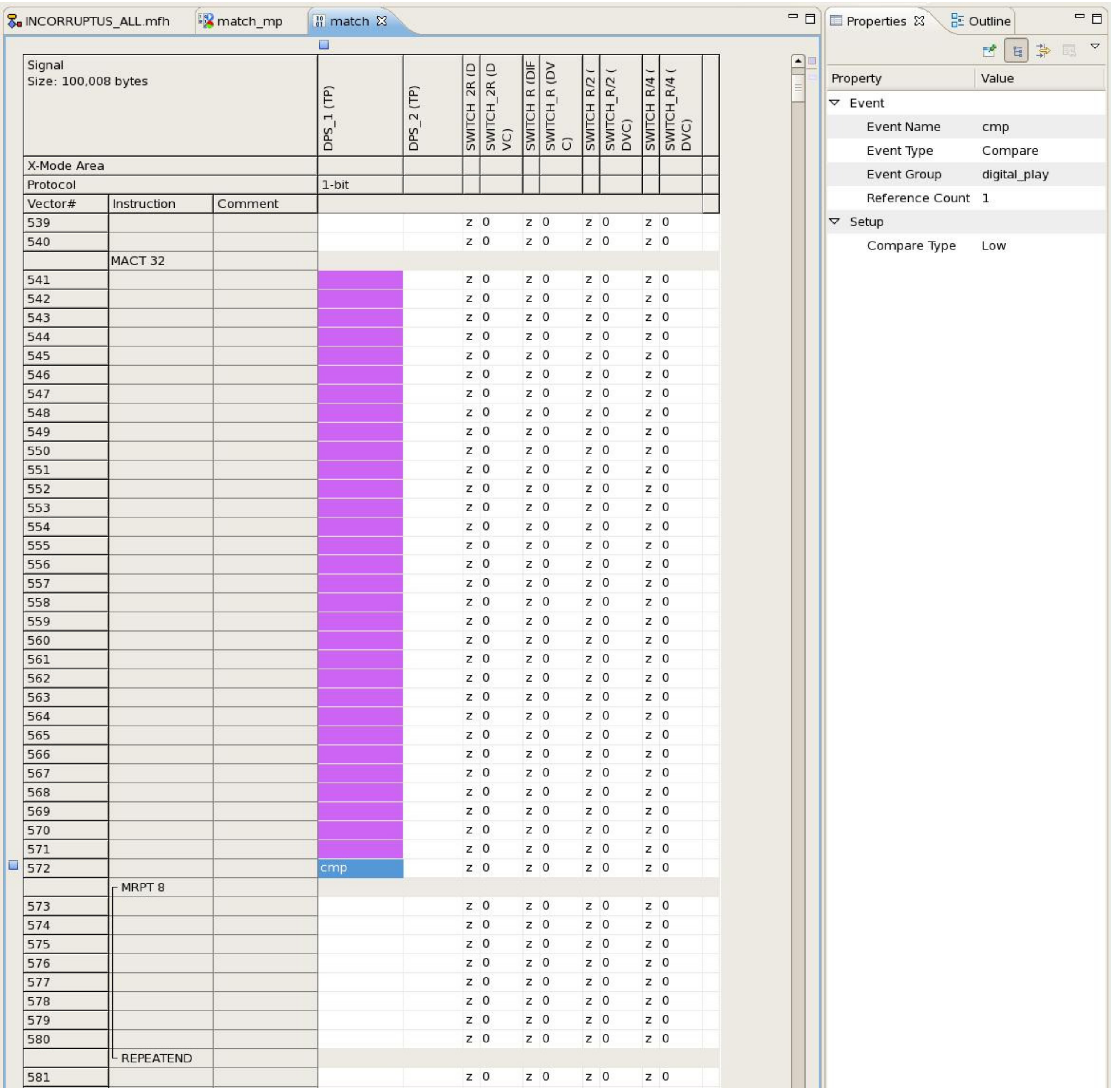


Figure 16: DPS128 participating in a match-loop; Compare event in pattern stream

2.5 Partial Ganging

Ganged operation is defined for applications that require more current than the single channel can provide. To setup a gang multiple DPS channels are connected in parallel.
Connecting multiple channels together increases the available current, both minimum and maximum limit. On the other hand the measure accuracy gets reduced. For both cases the effect is usually multiplied by the number of ganged channels. A gang is global to the test program.

In order to reduce the minimum clamp limit and/or improve the accuracy, it is possible to temporarily detach DPS128 channels from a gang (partial ganging).

In the force operation the number of effectively ganged channels is dynamically configured according to the value of the programmed voltage range and current range.
In the measure operation the number of effectively ganged channels is dynamically configured according to the limits of the current measurement.

2.5.1 Force Operation

For force operations the application focus of a gang is in providing more current than a single channel is able to provide. Setting up a gang increases the minimum and maximum current clamp limit by the factor of the gang.
Sometimes there are special test items that require a very low clamp current that might be below the minimum clamp limit of the gang. A typical use case is the power-short test.

The partial ganging mechanism is available for both instrument operation modes. The test system detaches as many channels as possible from the gang, and leaves only the required active channels in the gang which are necessary to achieve the required clamp current.
The amount of required channels depends on the actual voltage range as well as the current range. The voltage range is used to determine the maximum allowed current of one DPS channel at this voltage. The clamp range value is then divided by the maximum current value. The result is the number of required channels.

An explicit disconnect-connect sequence to enter and disconnect to exit is required.

Below is one example showing a gang setup for 5 channels, but only 3 are finally ganged.

- **DPS instrument mode:**
  To reduce the minimum clamp current in a ganged setup a special connect state "*ungang*" is provided.

  <u>Gang setup of 5 channels:</u>

	Name	Pol	C[uF]	HW	Comment	Tester Channel Site 1
1	VDD	POS	0.0	DCS-DPS128HC		20101-20105
2	VDDQ	POS	0.0	DCS-DPS128HC		20110

Partial gang of 3 channels via levels equation:

Implemented by the "*ungang*" connect state in the levels equation.
The values assigned to "*vout_frc_rng*" and "*iout_clamp_rng*" determine the number of channels to be kept in the gang. V-I curve dependencies need to be considered.

```
DPSPINS VDD
vout           = 2
vout_frc_rng   = 2.5
iout_clamp_rng = 2800
connect_state  = ungang
```

(2.8 A @ 2.5 V can be done with 3 DPS128 channels)

Also refer to TDC topic 130681.

- **PMU instrument mode:**
  To reduce the minimum clamp current in a ganged setup it is possible to combine the connect execution mode with the un-gang measurement mode (without actually doing a measurement).

Gang setup of 5 channels:

	Name	No	Type	Mode	HW	Comment	Tester Channel
							Site 1
1	VREF		dc	SIG	DCS-DPS128HC		42501-42505

Partial gang via SPMU_TASK Testmethod API:

Implemented by the combination of *FORCE_ON*" execution mode and "*VFIM_UNGANGED*" *or* "*IFVM_UNGANGED*" measurement mode.
The values assigned for "*vForceRange()*" and "*iForceRange()*" determine the number of channels to be kept in the gang. V-I curve dependencies need to be considered.

```
SPMU_TASK connect;
connect.pin("GANG_REF").mode("VFIM_UNGANGED").relay("NT")
       .vForceRange(2.5 V).vForce(2 V)
       .iForceRange(2800 mA).clamp(2500 mA)
       .min(0 mA).max(1 mA);    // dummy, as pure connect, no measurement
connect.execMode(TM::FORCE_ON).execute();
```

(2.8 A @ 2.5 V can be done with 3 DPS128 channels)

2.5.2 Measure Operation

For measure current operations the application focus is in increasing the measurement accuracy. Therefore, some partial ganging mechanism is available to temporarily un-gang channels.

For both instrument modes the measurement operation is triggered by Testmethod API. In both cases the values assigned for the "*max()*" and "*min()*" pass/fail limits as well as the clamp limit determine the measurement range. The values also define the number of channels to be temporarily detached respectively remained in case of an un-ganged current measurement.

The following rule is applied:
(No. of remaining ganged pins) (clamp limit) / (total defined ganged pins) >= pass/fail limit*

- **DPS instrument mode:**
  Implemented by "*TM::MEASURE_CURRENT_UNGANG*" mode option of `DPS_TASK` API.
  The clamp limit is defined in the level equations ("ilimit").

```
DPS_TASK measure;
measure.pin("VDD").measurementMode(TM::MEASURE_CURRENT_UNGANGED)
        .min(100 mA).max(200 mA);
measure.execMode(TM::PVAL).execute();
```

  Refer to TDC topic 55665 for more details.

  Alternatively it is possible to utilize the *"ungang"* connect state to limit the amount of ganged channels to the minimum required. This requires at least one disconnect - connect sequence upfront of the test run.
  Note that this is not a dynamic current limit change (measure accuracy change) just for the measurement part of the test, but of global nature to the test.

- **PMU instrument mode:**
  Implemented by *VFIM_UNGANG"* mode option of `SPMU_TASK` API.
  The clamp limit is defined by the "*clamp()*" parameter.

```
SPMU_TASK measure;
measure.pin("GANG_REF").mode("VFIM_UNGANGED").relay("NT")
        .vForceRange(3 V).vForce(2.5 V)
        .clamp(4000 mA)            // 4A @ 3V vRange --> gang of 5
        .min(100 mA).max(200 mA);  // go to the single channel
measure.execMode(TM::PVAL).execute();
```

Note: The un-gang mechanism works for "code based" static operation only. This is related to the non-sequencer controlled relays (force / sense).

2.6 Asymmetric Clamp Current

DC Scale DPS128 enables different clamp current source / sink limits in DPS as well as PMU instrument mode.

- **DPS instrument mode:**
  "`ilimit_source`" and "`ilimit_sink`" in levels equation, i.e.

```
DPSPINS VDD
iout_clamp_rng = 1000    # current range the clamp operates in
ilimit = 750             # sets both source and sink (compatible)
ilimit_sink = 200        # sets only sink current
ilimit_source = 750      # sets only source current
```

- **PMU instrument mode:**
  Enhanced *"`clamp()`"* method of `SPMU_TASK` Testmethod API, i.e.

```
SPMU_TASK dc;
dc.pin("VREF").iForceRange(1000 mA);
dc.pin("VREF").clamp(750 mA);           // sets common sink/source
dc.pin("VREF").clamp(200 mA, 750 mA);   // sets individual sink/source
```

This capability is available for all DC Scale cards and is enabled with the release of DC Scale DPS128.

2.7 Voltage Ramp Control

In DPS instrument mode DC Scale channels are able to setup individual voltage ramps. This allows specifying setup times and slew rates separately for rising and falling DPS voltage transitions.
The voltage ramps can be used for all DPS voltage changes, regardless of whether they are caused by connect, disconnect, a change within the same DPS set, or a switch between DPS sets.
Ramp control does not apply to Dynamic DC force actions. In this case dynamic actions need to be used to program a ramp.

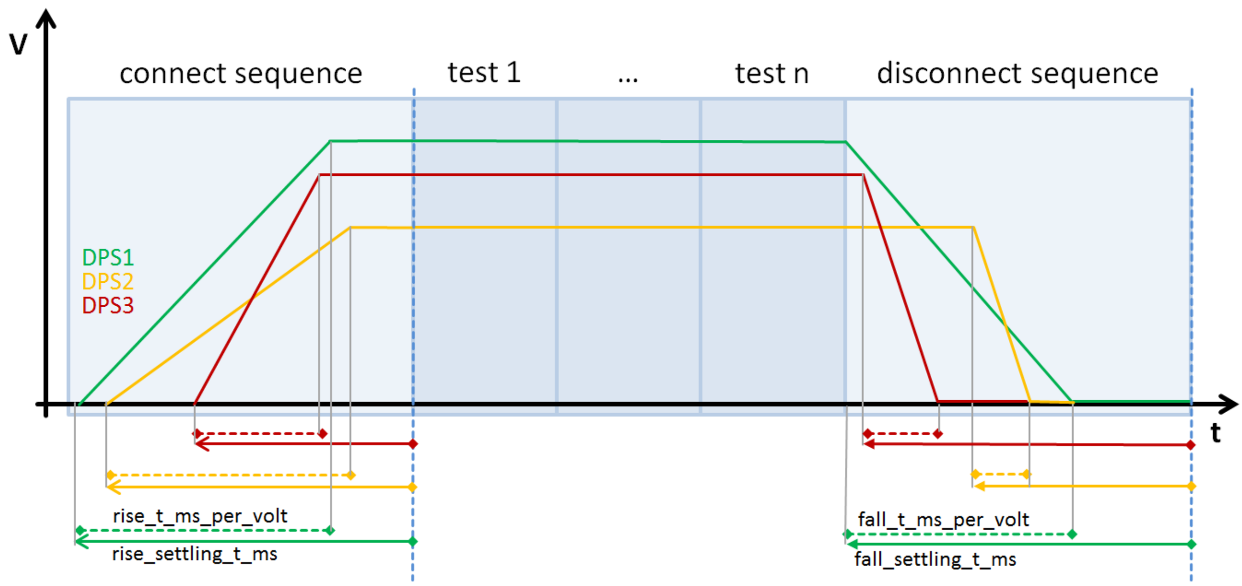


Figure 17: DC Scale voltage ramp concept

Voltage ramp control requires adding certain key-words to the equation block of the level setup.

Key-words used for connect and disconnect sequences:

- *vout_rise_t_ms_per_volt* the slew rate for the rising edge in ms/V
- *vout_fall_t_ms_per_volt* the slew rate for the falling edge in ms/V
- *vout_rise_settling_t_ms* the settling time for rising edge in ms
- *vout_fall_settling_t_ms* the settling time for falling edge in ms

Key-words used for “connect-to-connect” sequences, that is for level changes caused by a reprogramming of the output voltage while keeping the DPS set, or by switching to another DPS set while the DPS is connected:

- *vout_rise_t_ms_per_volt_c2c* the slew rate for the rising edge in ms/V
- *vout_fall_t_ms_per_volt_c2c* the slew rate for the falling edge in ms/V
- *vout_rise_settling_t_ms_c2c* the settling time for rising edge in ms
- *vout_fall_settling_t_ms_c2c* the settling time for falling edge in ms

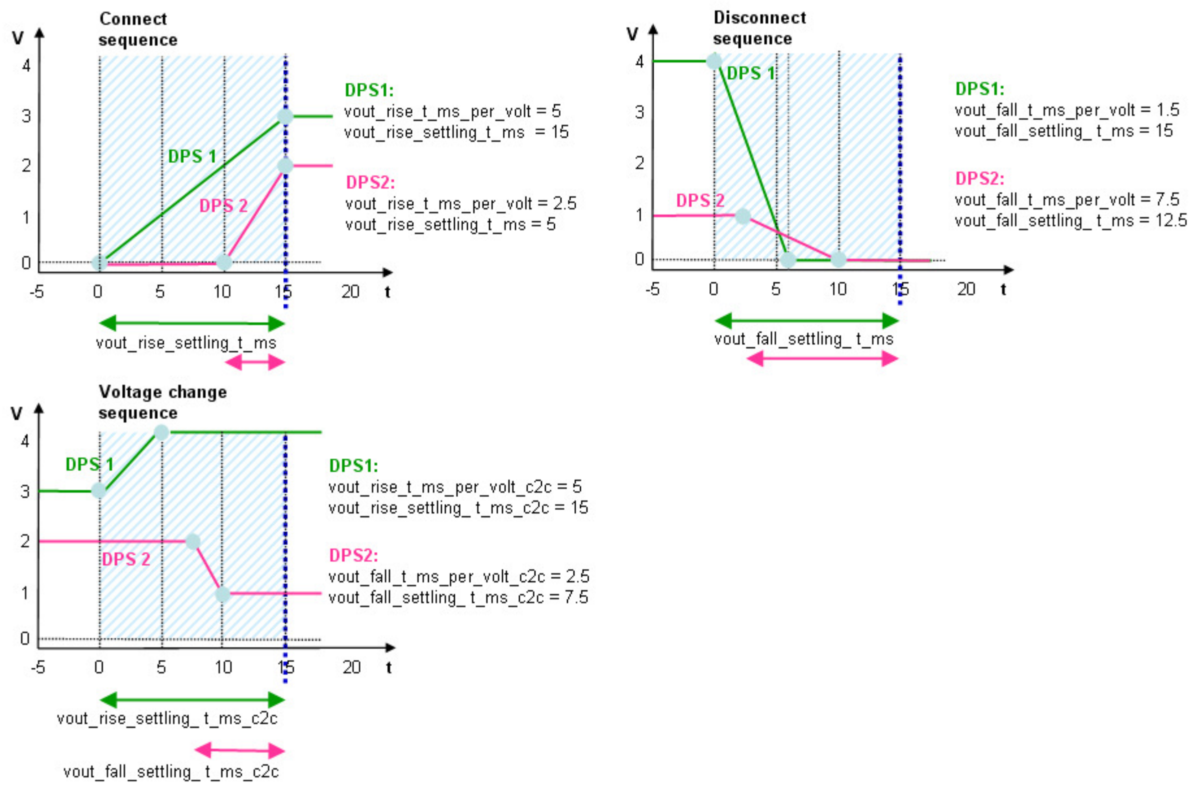


Figure 18: Voltage ramp details

Please refer to TDC topic 125205 for more details on DPS voltage ramp application.

2.8 High Impedance Measurement Mode

DPS128 provides a high impedance measurement mode (“Voltmeter Mode”) that implements a low current, (high) voltage type of measurement. In “Voltmeter Mode” the force line is disconnected while the sense line is still connected.

This mode requires an explicit disconnect-connect sequence to enter and disconnect to exit. It is available for “code-based” static operation as well as “pattern-based” dynamic operation.

- **DPS instrument mode:**
  Voltmeter Mode enabled via special connect state (“*himp*”) in the DPS section of the levels equation.

```
DPSPINS  VDD
vout             = 2     #[V]
vout_frc_rng     = 2.5   #[V]
iout_clamp_rng   = 200   #[mA]
ilimit           = 100   #[mA]
t_ms             = 1     #[ms]
offcurr          = act   # disconnect state
connect_state    = himp  # connect state: HIZ measurement mode
```

- **PMU instrument mode:**
  Voltmeter Mode enabled via mode option “*VM_HIZ*” of SPMU_TASK API.
  By default it will operate in the highest voltage range and smallest current range.

```
SPMU_TASK himp;
himp.pin(sigPin).mode("VM_HIZ").relay("NT")
    .min(0 mV).max(50 mV).settling(1 ms);
himp.execMode("PVAL").execute();
```

  For pattern controlled execution the pin needs to be connected with execMode option “*FORCE_ON*” before the pattern run and disconnected with “*FORCE_OFF*” after the pattern run.

Please note, that DPS128 does not implement a dedicated HIZ relay between force and sense line. Also refer to the chapter “High Impedance Measurement – Voltmeter” (Compatibility section).

2.9 Cap-less Mode

DPS128 allows for cap-less operation that is defined as working with some bypass capacitor network ≤ 10nF.

From application point of view cap-less mode is focused on VI style operation and will influence DPS128 internal voltage regulation loop as well as speed up general performance, i.e. slew-rates, settling times.

VI (and cap-less) style operation is supported when working in PMU instrument mode and available for both types of channel (HC and HV).

Operating in DPS instrument mode some minimum capacitance of > 1µF is required.

Load capacitance range	
Minimum required external capacitance with stability maintained	1nF to 10nF [1)]
Maximum supported external capacitance with stability maintained	470 µF per channel (n x 470 µF up to 5 mF if ganged) [2)]
Required external capacitance range for DPS style operation	1 - 100 µF per channel [2)]

Figure 19: DPS128 capacitance requirements (HC type); TDC topic 141967

Cap-less operation (≤ 10nF) needs to be defined in Smartest to allow for proper dynamic regulation of the DPS channel. This requires setting the load capacitance value to '0'. Any value > '0' will be ignored, i.e. Smartest selects the appropriate performance range automatically.

- **DPS instrument mode (not covered by specification):**
  Cap-less mode enabled via "C [µF]" column in pin configuration editor

*DPS Pins

	Name	Pol	C[uF]	HW	Comment	Tester Channel Site 1
1	VDD1	POS	0.0	DCS-DPS128HC	cap-less	42511
2	VDD2	POS	1.0	DCS-DPS128HC		42510

- **PMU instrument mode:**
  Cap-less mode enabled via "`loadCap()`" method of `SPMU_TASK` API.

```
// voltage force connect: force 2.5 V in 3 V voltage range
// 200 mA current range with clamp at 100 mA sink / 150 mA source
SPMU_TASK connect;
connect.pin(sigPin).mode("VFIM").relay("NT")
        .vForceRange(3 V).vForce(2.5 V)
        .iForceRange(200 mA)
        .min(0 mA).max(1 mA)     // dummy, as no measurement
        .clamp(100 mA, 150 mA)  // for different sink/source limits
        .loadCap(0);            // 0: cap-less mode
connect.execMode("FORCE_ON").execute();
```

Note: Switching between cap-less and non-cap-less mode is not possible while connected.

2.10 Profiling

DC Profiling allows for periodic measurements in the background. Using profiling the user can easily record the voltage or current curve of the actual patterns. This is extemely helpful especially for debugging use cases or finding the correct current measurement point for IDD/IDDQ.

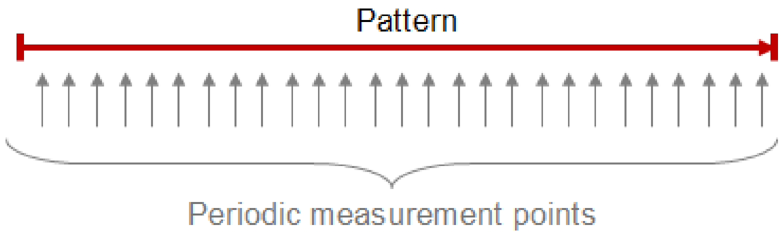


Figure 20: Profiling: Periodic measurement sequence

The maximum profiling sample rate is 50ksps for all channels in parallel. It allows for 100ksps rate on a single channel per logical board (exclusive).

Profiling is controlled by a special Testmethod API “`DC_PROFILING`” and / or via Hardware Monitor directly.

See “Testmethod Programming” for more details on Profiling Testmethod API.

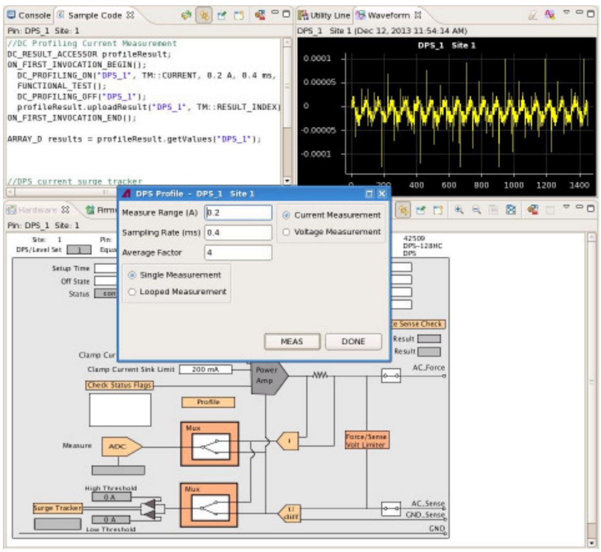


Figure 21: Profiling via Hardware Monitor

Profiling is based on pattern execution; therefore, it is mandatory to enable testpoints in the “.technology” file, even if it does not consume any of them.

Please note that enabling testpoints makes the DPS128 to operate in a dynamic mode for the whole device setup. This has certain consequences, as described in chapter “Compatibility” and “Dynamic Operation”.

2.11 AWG / Digitizing Capability

When setting up DPS128 in dynamic operation it is possible to perform basic mixed signal tasks. This is about sourcing waveforms (incl. multi-tone) and digitizing them.

To perform mixed-signal tasks DPS utilizes the standard DPS Testmethod API for dynamic operation, as well as the standard mixed-signal Testmethod API to generate waveforms.
For digitizing it is possible to use the existing mixed-signal DSP-API to perform static or dynamic parameter calculations.

A ramp example can be found in the Testmethod Programming chapter “Dynamic Current”.

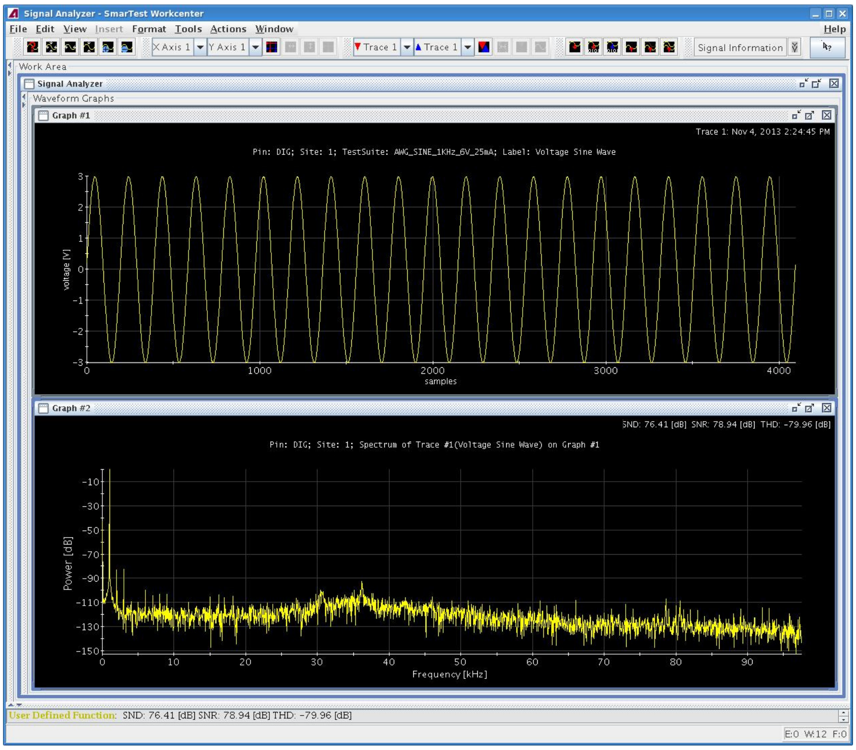


Figure 22: 1 kHz / 6 Vpp sine generated by DPS128 HC (no filter)

Due to its nature of a DPS the resource is limited in sample rate and bandwidth, which are 200ksps/5kHz for AWG operation and 90ksps/20kHz for Digitizer operation.

It does not provide dedicated filter settings on board; therefore, care has to be taken during loadboard design.

2.12 DC Scale Limits

DC Scale Limits is a new safety feature for all DC Scale power supply cards. The Limits protect the device against too high or low voltage or current levels.

The intention is to not allow any voltage and current over-programming by the user. The Limits cannot be exceeded and are of global nature.

The Limits are integrated to the pin configuration setup and can be defined per pin.

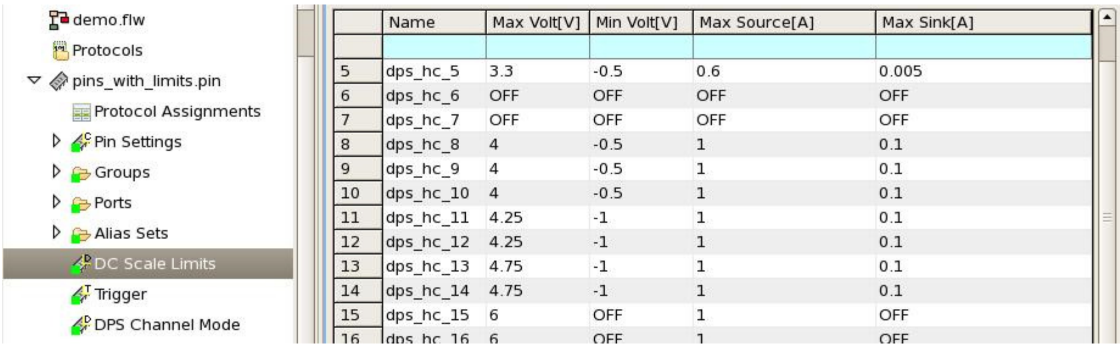


Figure 23: DC Scale Limits

Please refer to the TDC topic 142044 for more details.

3 Power Budget

DC Scale DPS128 has a strong focus on multi-power domain applications, as well as, high multisite applications. Therefore, the overall power budget needs to be seriously considered.

3.1 V-I Curves and Board Power Restrictions

The V-I curves of the two types of DPS128 channels are quite different. Despite the fact of different voltage ranges and less current capability for HV type of channels, the HV type of channels do not have any current limit to voltage dependency over their voltage operating range as there is for HC type. In other words a HV type of channel can provide its maximum current limit for all possible voltages over the whole operating range.

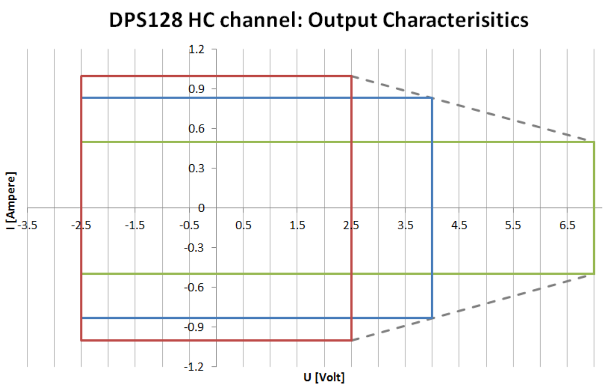


Figure 24: HC type of channel output characteristics

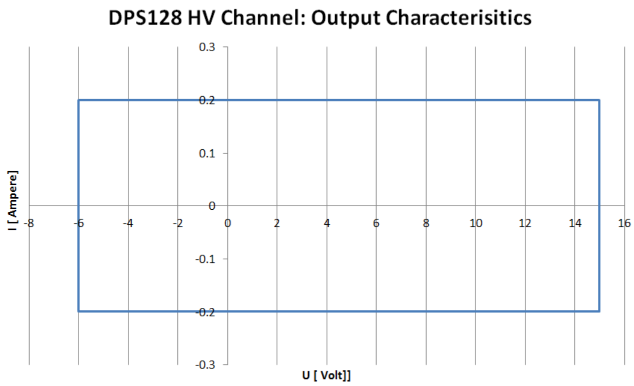


Figure 25: HV type of channel output characteristics

Note that for HC type of channels the maximum current limit depends on the voltage range. For a specified voltage range of 2.5 V the maximum current limit is 1 A, while for a specified voltage range of 7 V the maximum current is 500 mA. For voltage ranges between 7 V and 2.5 V the maximum current limit decreases in a linear manner, i.e. for a specified voltage range of 4 V the maximum current is 0.833 A.

To achieve 3 A @ 4V it is required to gang

- 4 channels, if voltage range is set to 4 V
- 6 channels, if voltage range is set to 7 V (default)

Please refer to chapter “Voltage Ranges” for more details.

What also needs to be taken into consideration is that the V-I curve cannot be mapped one-to-one to a complete DPS128 board, i.e. the maximum current limit of a channel cannot be applied to all channels of a board in parallel. This restriction is driven by the overall available power budget of a board.

Differences in maximum current source / sink capabilities per board are listed below:

E8013CS - DPS128-HC, HC type of channel, 128 channels:

- maximum current per channel is 1 A @ 2.5 V; 0.833 A @ 4 V; 0.5 A @ 7 V
- 66 A / 58 A @ 2.5 V maximum source / sink current per board
- 1 A / 0.879 A @ 2.5 V simultaneously on 66 channels (if others at lowest current range)
  (66 channels → 33 channels per power domain)
- 0.480 A / 0.411 A @ 2.5 V simultaneously on 128 channels
- 0.346 A / 0.405 A @ 4 V simultaneously on 128 channels
- 0.209 A / 0.394 A @ 7 V simultaneously on 128 channels

E8023CSH - DPS64-HC, HC type of channel, 64 channels:

- maximum current per channel is 1 A @ 2.5 V; 0.833 A @ 4 V; 0.5 A @ 7 V
- 64 A / 64 A @ 2.5 V maximum source / sink current per board
- 1 A / 1 A @ 2.5 V simultaneously on 64 channels
- 0.833 A / 0.833 A @ 4 V simultaneously on 64 channels
- 0.5 A / 0.5 A @ 7 V simultaneously on 64 channels

E8013CSW - DPS128-HV, HV type of channel, 128 channels:

- maximum current per channel is 0.2 A
- 13.2 A / 24.4 A maximum source / sink current per board
- 0.089 A / 0.188 A simultaneously on 128 channels

E8013CSV- DPS128-HC/HV, mix of HC and HV type of channel, 64 channels each:

HC portion:

- maximum current per channel is 1 A @ 2.5 V; 0.833 A @ 4 V; 0.5 A @ 7 V
- 62 A / 54 A @ 2.5 V maximum source / sink current per board (only HC channels used)
- 0.983 A / 0.849 A @ 2.5 V simultaneously on 64 channels (only HC channels used)
- 0.726 A / 0.833 A @ 4 V simultaneously on 64 channels (only HC channels used)
- 0.465 A / 0.5 A @ 7 V simultaneously on 64 channels (only HC channels used)

HV portion:

- maximum current per channel is 0.2 A
- 12.8 A / 12.8 A maximum source / sink current per board (only HV channels used)
- 0.2 A / 0.2 A simultaneously on 64 channels (only HV channels used)

Please refer to the chapter “Power Budget Decision Model” for more information on power budget related dependencies like programmed current limits and voltage ranges.

3.2 Power Domains and Pogo Assignment

The DPS128 board is supplied by two power domains with a limited power budget. The software limits the programmed clamp current of all channels, so that the power budgets of the power domains are not exceeded. The following figure shows the physical locations of the channel modules and the power domains:

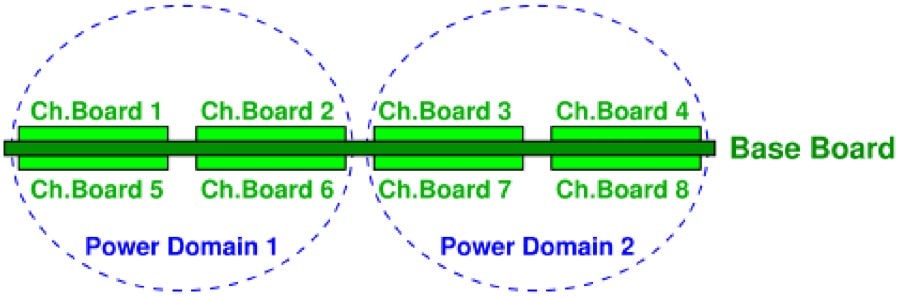


Figure 26: DPS128 power domains

This gives the following assignment of channels to the power domains:

Power domain:	Channels:
1	• 1 ... 32 (channel modules 1 and 2) • 65 ... 96 (channel modules 5 and 6)
2	• 33 ... 64 (channel modules 3 and 4) • 97 ... 128 (channel modules 7 and 8)

Figure 27: Power domain to channel (module) mapping

Each domain is able to deliver half of the overall current budget. From the application point of view it is important to distribute the device current requirements based on the power domain mapping otherwise serious power restrictions might apply.

The above leads to the following pogo-mapping (using E8023PA pogo cable):

DPS128/DPS64 variant:	Channel modules	Channel modules type	Channels	Used pogo pin blocks
DPS128-HC (E8023CS)	1 - 8	HC	1 - 128	A - H
DPS64-HC (E8023CSH)	1 - 4	HC	1 - 64	A - D
	5 - 8	empty	none	E - H
DPS128-HC/HV (E8023CSV)	1 - 4	HC	1 - 64	A - D
	5 - 8	HV	65 - 128	E - H
DPS128-HV (E8023CSW)	1 - 8	HV	1 -128	A - H

Figure 28: Power domain to pogo mapping (E8023PA)

Please refer to the chapter “A2 – Pogo Cable Channel Mapping” for more details on the pogo-mapping / power-domain mapping for the various cable derivate.

3.3 Voltage Ranges

Voltage ranges are implemented to utilize the maximum specified output current per channel as well as to effectively maximize the available current budget per board.
Please note that the voltage ranges are not linked to any electrical performance or accuracy specification and are only applicable to HC type of channels.

1. Maximum output current.
   The maximum clamp current limit depends on the voltage range. For a specified voltage range of 2.5 V the maximum current is 1 A, while for a specified voltage range of 7 V the maximum current is 500 mA.
   For more details refer to chapter “V-I Curves and Board Power Restrictions”

2. Power budget optimization.
   Using voltage ranges will minimize the current dissipation, which on the other hand increases the effectively usable current budget per power domain.
   SmarTest internally calculates a certain current margin in dependency to the programmed clamp current and programmed voltage range. By limiting the voltage range the available current margin will increase. The amount of effective power gain depends on multiple factors and is built up in a very complex formula that cannot be described here.

Without specifying any voltage range the HC type of channel always works in the maximum voltage range of 7 V (DPS32 compatible). This will significantly limit the effective current budget of a power domain, as the SmarTest internal power budget calculations are based on a potential force voltage of 7 V (that may never be programmed / applied).
Therefore, in order to optimize the overall power usage of the board, it is mandatory to setup voltage ranges for all channels, as soon as some of them are operating in the 1 A current range.

The voltage range can be programmed to any value (no dedicated range), while all values below 2.25 V will be auto-corrected to 2.25 V (minimum voltage range for 1 A current range operation).
In total there are 256 internal range steps from 2.25 V up to 7 V. This gives a resolution of ~18 mV per step. Any programmed value will therefore be mapped to the next higher range based on the resolution.

Limiting the voltage range can be done by:

- DPS operation mode:
  “*vout_frc_rng*” keyword in level equations, i.e.

```
DPSPINS VDD
vout          = 1.8
vout_frc_rng  = 3.5      #max bump voltage
```


- SIG operation mode:
  "*vForceRange*" API of `SPMU_TASK`, i.e.

```
SPMU_TASK dc;
dc.pin("VREF").vForce(1.8 V).vForceRange(3.5 V);
```

Note:

- The voltage range should define the maximum voltage that is applied while the DPS is in connected state, i.e. any dynamic actions that may "over-program" the force voltage need to be considered.
- Switching the voltage force range requires a disconnect sequence. From a test time point of view no spec should be used, but a maximum value that is valid for all relevant specs and equations.
- The HV type of channel does not implement voltage ranges. It operates solely in the 15 V range fix

3.4 Power Budget Decision Model

To prevent any likelihood of an over-power shutdown of the DPS card, SmarTest will limit the overall power usage and issue an error message if it is violated.
The decision in general is based on the programmed current clamp limit of the channels, with the HC type also taking care of the programmed voltage range.

For the DPS instrument mode the parameters are defined in the level equations ("*ilimit*", "*vout_frc_rng*") and therefore the decision is based on the level primary set and will be triggered during its activation.
For example, the below DPS128-HC setup will trigger the error message during primary set activation, as it is impossible to source / sink 1 A @ 2.5 V on all 128 channels in parallel.

```
DPSPINS  all_128_channels
vout_frc_rng    = 2.5        # [V]
vout            = 2.5        # [V]
iout_clamp_rng  = 1000       # [mA]
ilimit          = 1000       # [mA]
```

For PMU instrument mode the current clamp limit and voltage range is typically specified during the connect sequence of the instrument.
Please refer to chapter "DSP128 Setup & Programming - PMU Instrument Mode" for more details.

In the "pattern-based" dynamic mode SmarTest will also analyze the programmed test points for the defined current clamp limits (current force) on a per pattern base. Based on this an error will be triggered during pattern activation. There is no check for the voltage range, as it cannot be changed during the pattern execution.

The power characteristics are different for the HC and HV channel version:

HC type of channels:

- the most dominant factor is the specified voltage range with a minimum value of 2.25 V
- current sink is a more dominant factor than current source for voltage ranges < 3.3 V
- current source is a more dominant factor than current sink for voltage ranges > 3.3 V
- no limitations when all channels operate in current ranges < 1 A

HV type of channels:

- no dependency to voltage ranges; only a single range is available
- current source is the dominant factor

Please refer to chapter "Asymmetric Clamp Current" for more detail on specifying separate clamp limits.

3.5 Power Budget Calculator

A power budget calculator is available and covers following use cases:

- migrate from DPS32 cards to the DPS64 and DPS128 cards
- configure a new DPS128 test system to test a new device
- check and enhance an existing DPS128 test system to test a new device
- calculate the power consumption of predefined DPS128 cards

Refer to "DC Scale DPS128 Power Calculator" TDC topic 147808 for more details.

3.6 Disconnect State

To optimize the internal power budget any channel is programmed to the 25 mA current range, as soon as the channel enters certain disconnect states. Re-programming does not happen in the case where the actual specified current range is already smaller than the 25 mA range.

Disconnect states that trigger an internal reprogramming of the current range are:

- "*min*"
- "*act_min*"
- "*open*"
- "*act_open*"

Note that the "*act*" disconnect state is not affected, i.e. it keeps the current range as specified in the DPS section of the level equations.

For a description of the disconnect states refer to TDC topic 16887; parameter: "*offcurr*".

When connected the channels are programmed to the maximum current range unless otherwise specified (DPS32 compatible).

3.7 DC Scale DPS32 / MSDPS Replacement

In the context of power budget and maximum current force capability it is recommend using DPS64 HC for “pure” replacement actions.

The DPS64-HC and according pogo cable matches or even exceeds the maximum current force capability of DPS32 and MSDPS, while DPS128 HC does not.

The following card/pogo combinations are recommended:

DC Scale DPS32:

- DPS64 HC with E8023PD pogo cable
- 32 channels, DPS32 compatible pogo layout (2 IO blocks)
- 2 A @ 2.5 V max per channel
- Replaces 1x DSP32

MSDPS:

- DPS64 HC with E8023PF pogo cable
- 16 channels, MSDPS compatible pogo layout (2 MSDPS blocks, QUL-like)
- 4 A @ 2.5 V max per channel
- Replaces 2x MSDPS

4 Current Ranges & Current Force/Measure Scenarios

4.1 Combined Force and Measure Ranges

DC Scale DPS128 implements multiple current force ranges, as well as measure ranges from 1 A down to 12.5 µA.

From a hardware perspective DPS128 does not have separate force and measure ranges, but the software will still show separate ranges to the user. The implementation of this kind of combined hardware ranges is different to any existing DC Scale cards as well as MSDPS and requires an auto-ranging use model for the measurement case.

The following table shows the dependency between the force ranges and the (virtual) measure ranges. All supported range combinations are listed.

Current measure range	Current force range										
	12.5 µA	25 µA	125 µA	250 µA	1.25 mA	2.5 mA	12.5 mA	25 mA	100 mA	200 mA	1 A
±12.5 µA	ok		ok		ok		ok		ok		ok
±25 µA		ok		ok		ok		ok		ok	
±125 µA			ok		ok		ok		ok		ok
±250 µA				ok		ok		ok		ok	
±1.25 mA					ok		ok		ok		ok
±2.5 mA						ok		ok		ok	
±12.5 mA							ok		ok		ok
±25 mA								ok		ok	
±100 mA									ok		ok
±200 mA										ok	
±1 A											ok

Figure 29: HC type of channel current range combinations

If a force range / measurement range combination is set that is not listed, it is still possible to execute and measure it. In this case the measurement range is internally auto-corrected to the next higher measurement range. For example, programming the measurement range to 125 µA when in 200 mA force range switches the DPS to the 250 µA measurement range (HC type of channels).

Current measure range	Current force range				
	25 µA	250 µA	2.5 mA	25 mA	200 mA
± 25 µA	ok	ok	ok	ok	ok
± 250 µA		ok	ok	ok	ok
± 2.5 mA			ok	ok	ok
± 25 mA				ok	ok
± 200 mA					ok

Figure 30: HV type of channel current range combinations

4.2 Current Measurement Scenario

4.2.1 Auto-Ranging Use Model

Auto-ranging is defined as an automated down ranging based on a user defined target range. The ranging is executed in a sequential way, range by range, i.e. directly jumping to the target range is not possible.

DC Scale DPS128 implements an auto-ranging use model for the reasons below:

1. Hardware does not allow to have a different current measure range than the actual current force range, but measurements require to down-range to achieve the required measurement accuracy
2. It is not allowed to explicitly change the current force range without a disconnect sequence

Note that for DPS128 auto-ranging is not about automatic measure range selection based on the measured value, but about an automation of the step-by-step down-range sequence without any disconnect.

The down-ranging sequence is encapsulated within a special algorithm. Within this sequence the ranges get changed step by step without any disconnect in-between. The measurement range will only be changed if the actual flowing current is small enough, i.e. no current limiting will occur when switching to the next lower range. The internal limit set by SmarTest is 90% of the current limit of the next lower range.
Note that it is only allowed to set a measurement range that is lower than the force range.

There is a status flag available indicating whether down-ranging succeeded or not, i.e. the intended target range was set and no clamping occurred during the sequence. The flag can be accessed by "`DPS_AUX_TASK`" Testmethod API.
Refer to "Testmethod Programming" for more details.

The measurement itself is always executed in the range that is actually set, no matter the ranging succeeded or not. In the case of a down-range failure the measurement will be executed in the last (valid) range set by the down-ranging sequence.

After the measurement task the original force range is restored by SmarTest. This happens automatically to avoid potential current limit issues ("constant current scenario") for any follow-on tests. This is especially important for "pattern based" dynamic execution in case the pattern stops execution unexpectedly because of some fatal error.

In total the measurement sequence for DPS128 consists of 3 sub-tasks:

- Set measurement range (incl. auto-ranging)
- Measurement
- Restore force range

In “code-based” static operation the measurement sequence is executed at a glance, which is completely controlled by SmarTest. This also includes the minimum program, settling and sample times required for the measurement sequence. However, it is possible for the user to specify additional settling times if required.

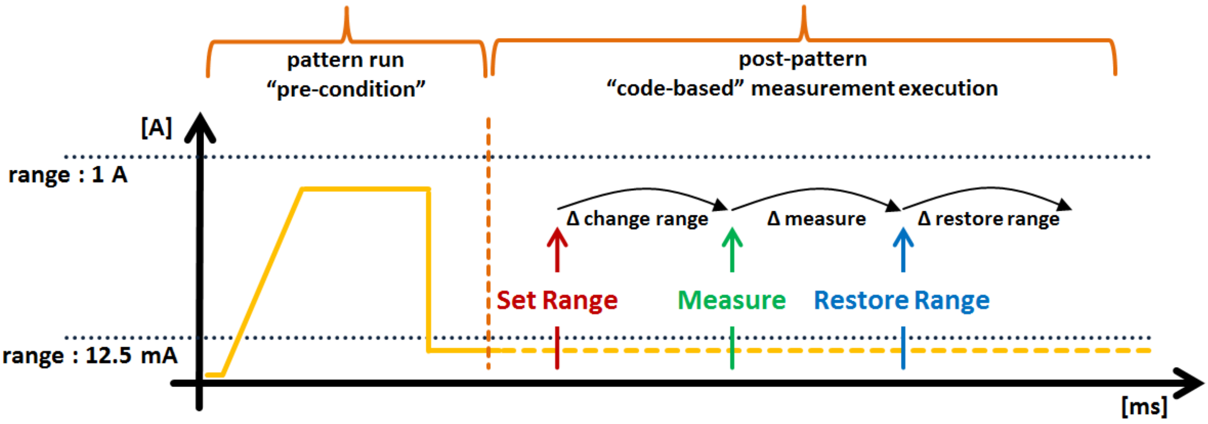


Figure 31: "Code-based" static IDDQ measurement sequence example

In “pattern-based” dynamic operation the set range task and the measure task need to be explicitly programmed by the user inside the pattern, while the restore task is triggered by SmarTest automatically after the pattern run.
Required wait times for set range and measure task need to be explicitly programmed to the pattern by the user. For set range task this can be done with the means of pad vectors or “*WAIT*” sequencer instruction, while for the measure task, pad vectors are required for result processing.

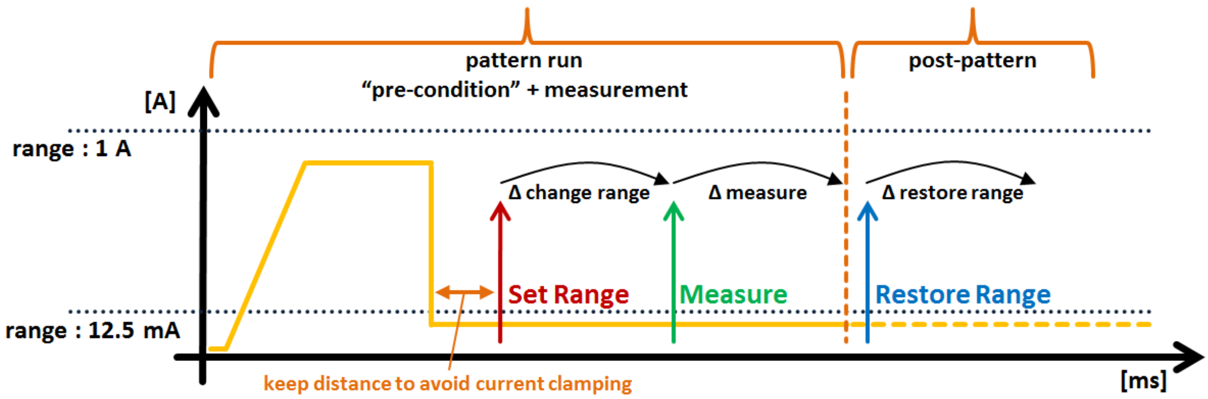


Figure 32: "Pattern-based" dynamic IDDQ measurement sequence example

In case of additional pattern tasks after the measurement that require some current flow higher than the maximum limit of the actual selected force/measure range combination it is required to execute a restore task. The task should be triggered after the measurement to avoid “constant current” scenarios later in the pattern. The restore task has to be explicitly programmed by the user in this case.
Required wait times, especially the restore range settling needs to be programmed to the pattern by the user with the means of pad vectors or “*WAIT*” sequencer instructions.

One potential use case of above requirement is a measurement burst, i.e. multiple patterns concatenated to one big burst pattern.

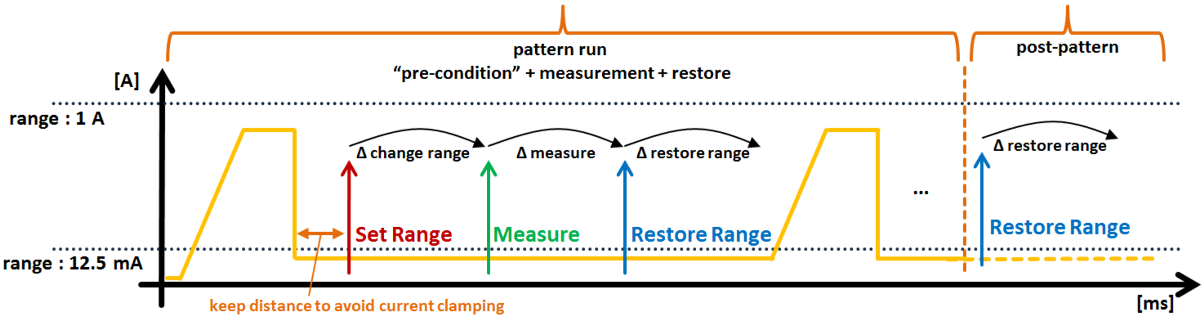


Figure 33: "Pattern-based" dynamic burst IDDQ measurement sequence example

Note that the restore task is always triggered after the pattern run (in case the range was changed before); even if the user has already included a restore task in the pattern.

Static and Dynamic Testmethod examples can be found in the Chapter "Testmethod Programming".

4.2.2 Settling Times in "Pattern-Based" Operation

In "pattern-based" dynamic operation the user needs to consider the range change wait times when setting up test points in the pattern. This is typically done by adding pad vectors or applying the sequencer instruction "Wait" after any range change event.

Wait times defined for DPS128 measurement ranging:

1. Range settling time.
   Required wait time to achieve the specified accuracy of the target range. This is defined by a hardware settling into the 1% range of the target range.
   It's not mandatory by the system, i.e. no warning or error will be issued in case if violated. However it is mandatory to achieve the specified accuracy.

2. Auto-range execution time.
   Required wait time to allow the auto-range algorithm to finish. This is a mandatory minimum time required by the system and an error will be reported if violated.
   For more information about the auto-range algorithm refer to chapter "Auto-Ranging Use Model"

The sum of above mentioned wait times defines the total wait time required by DPS128 range events. Potential device related settling times need to be added on top.

Range Settling Times in µs:

	To Range					
	1A*	200mA, 100mA*	25mA, 12.5mA*	2.5mA, 1.25mA*	250µA, 125µA*	25µA, 12.5µA*
Range Down		500	500	1320	10000	80000
Range Up	500	500	500	1000	8000	

Auto-Range Execution Times in µs:

From Range	To Range					
	1A*	200mA, 100mA*	25mA, 12.5mA*	2.5mA, 1.25mA*	250µA, 125µA*	25µA, 12.5µA*
1A*	0	20	290	700	1310	7320
200mA, 100mA*	10	0	20	440	1050	7060
25mA, 12.5mA*	10	10	0	20	640	6650
2.5mA, 1.25mA*	20	10	10	0	20	6040
250µA, 125µA*	20	20	10	10	0	20
25µA, 12.5µA*	20	20	20	10	10	0

Required Total Wait Times in µs:

From Range	To Range					
	1A*	200mA, 100mA*	25mA, 12.5mA*	2.5mA, 1.25mA*	250µA, 125µA*	25µA, 12.5µA*
1A*	0	520	790	2020	11310	87320
200mA, 100mA*	510	0	520	1760	11050	87060
25mA, 12.5mA*	510	510	0	1340	10640	86650
2.5mA, 1.25mA*	520	510	510	0	10020	86040
250µA, 125µA*	520	520	510	1010	0	80020
25µA, 12.5µA*	520	520	520	1010	8010	0

*: DPS128HC only

Down-ranging with change from Voltage to Current Measurement mode: Add 100µs to the execution and required wait times.
Same current range with change of measurement mode: Set required wait time to 500µs.

Figure 34: Required wait times for range change events in "pattern based" dynamic operation

Note that the range settling time is valid for a certain maximum capacity as documented in the specification. Any up-range event is only possible to the initial force range (restore task).

The range settling time can be omitted in case the accuracy does not care. In this case the required total settling time equals the pure auto-range execution time.
A typical use case for this scenario is a measurement burst (within a single sequencer run) that requires some repeating "set range – measure – restore range" approach, i.e. IDDQ burst. In this case the set range event will apply the total wait time, while the restore range event only cares for the auto-range wait time. This is possible as the responsibility of the restore event is to allow higher current flow, but not measurement accuracy.

Note that the automated restore range task at the end of the pattern run works like this.

4.2.3 Auto-Ranging & Ganged Channels

Auto-Ranging is not supported for ganged channel operation. This is due to the fact that ganged channels only operate in the highest current range (1 A for HC type of channels; 200mA for HV type of channels). No range selection is possible.

Measurement accuracy is based on the accuracy of the highest range multiplied by the amount of ganged channels (n). For HC type of channels this is ± (n x 1 mA +0.1 % of total current).

If the current requirement of the application is less than (n x 1A) the measurement accuracy can be improved by using the *"ungang"* connect state. This will only gang the minimum required amount of channels, which in turn will increase the measurement accuracy. If just one channel is left auto-ranging is applicable again.
Note that this alternative depends on application requirements and therefore cannot be applied in all cases. It also requires at least one disconnect – connect sequence before the pattern run.

Also refer to chapter "Partial Ganging"

4.2.4 Current Measurement Task Characteristics: "Set Measure Range"

- Auto-ranging approach with step-by-step down-ranging.
- User has to assure that the flowing current fits the new measurement range.
  Before the execution of an individual down-range step an internal measurement is done to ensure the clamp limit of the next lower range is not hit. If the limits were to be violated the ranging sequence would be stopped immediately and the actual range would be used to perform the measurement.
    - Stop condition is > 90% of the current limit of the next lower range in sequence
    - The well-known out-of-limit indication (±99999) will not be reported, as you are in valid range, just not the intended one
- Each down range step requires some internal wait time, but the overall range change wait time is dominated by the settling time of the target range. (Please refer to chapter "Settling Times in "Pattern-Based" Operation".)
- Ganged setups do not allow setting a current range other than the highest one (1 A for HC type of channels; 200mA for HV type of channels)

- **"Code-based" static operation:**
    - Task is automatically executed. This also includes the required wait times for range change and settling.
- **"Pattern-based" dynamic operation:**
  Event task to be added to the pattern by the user. User has to take care to add enough pad vectors between the set range task and the measure task to guarantee the range change has finished. This can also be accomplished by the usage of the "WAIT" sequencer instruction.
  Note that there should be at least 8 vectors after the event anchor and before the WAIT instruction. Please refer to table "Required wait times for range change events in "pattern based" dynamic operation".
  Also the task should be placed in pattern areas where the actual current flow fits the target current range. Otherwise ranging will not succeed.

4.2.5 Current Measurement Task Characteristics: "Measure"

- The required wait time of the measurement is determined by the amount of samples.
- Maximum number of samples (averaging) per measurement is 8192 (HC-type) or 4096 (HV-type).
- Required wait time for pure measure event, i.e. EVENT_IM, EVENT_VM:
  T = (6.5 + 5*n) µs ; n = samples
- Required wait time for combined force/measure event, i.e. EVENT_VFIM, EVENT_IFVM:
  T = (8.5 + 5*n) µs ; n = samples

- **"Code-based" static operation:**
  - Task is automatically executed. This also includes the required sample & hold time.
  - Number of required samples is set internally based on the specified result range by default, but can also be specified by the user.
- **"Pattern-based" dynamic operation:**
  - Event task to be added to the pattern by the user. User needs to take care to add enough pad vectors after the measure task to guarantee the required sample & hold time. There has to be at least 32 pad vectors to allow for the result processing.
    In difference to "Set Measure Range" event the "Wait" sequencer instruction is not supported in this case.
  - Default number of a sample is set to 1. User has to take care to specify enough samples to guarantee accuracy. (Refer to specification for more details).

4.2.6 Current Measurement Task Characteristics: "Restore Range"

- Range reset approach with step-by-step up-ranging.
- The individual range steps are programmed in a sequence without any wait / settling time for the individual steps, but the final range settling. (Please refer to chapter "Settling Times in "Pattern-Based" Operation".)

- **"Code-based" static operation:**
  - Task is automatically executed after the measurement. It is not possible to do continuous measure sequences within the same range, as there will be a restore after each measurement task.
- **"Pattern-based" dynamic operation:**
  - Task is automatically executed after the pattern run if the range was changed during pattern run.
  - [Optionally] Event task to be added to the pattern by the user based on pattern requirements (avoid "constant current" scenario when clamp is hit).
    User needs to take care to add enough pad vectors after the restore task event to guarantee the final range settling. This can also be accomplished by the usage of the "WAIT" sequencer instruction.
    Note that there should be at least 8 vectors after the event anchor and before the WAIT instruction.
    Please refer to chapter "Settling Times in "Pattern-Based" Operation".

4.3 Current Force Scenario

4.3.1 General Use Model & Considerations

To force current DC Scale DPS128 channels are utilizing their current clamp circuitry. This is the channels operate in constant current source mode. The direction of the current flow is decided by the potential difference between device output voltage and the programmed clamp voltage (force voltage). This is the same use-model as for the DC Scale DPS32.

- device output voltage > programmed clamp voltage → current flow into DPS
- device output voltage < programmed clamp voltage → current flow into device

For example, an IFVM task requires sourcing 10 mA to the device under test. The expected device output voltage is in the range of [1.25 V, 1.75 V].

In the DPS instrument mode the setup in level equation can be done as follows:

```
DPSPINS VDD
vout_frc_rng     = 3          # [V]
vout             = 2.5        # [V] clamp voltage
                              #     set to a value the DUT will never reach
iout_clamp_rng   = 25         # [mA]
ilimit           = 10         # [mA]
t_ms             = 1          # [ms]
offcurr          = act        # disconnect state
disable_const_curr_check      # surpress warning message for current force
```

As the clamp voltage of 2.5 V is always higher than the expected voltage, the current will flow from DPS to device. The voltage measurement should then read the value of the device output voltage.

In PMU instrument mode the setup can be done like this (“code based” static operation):

```
SPMU_TASK ifvm;
ifvm.pin(sigPin).mode("IFVM").relay("NT")
    .vForceRange(3 V).clamp(2.5 V)      // the clamp voltage
    .preCharge(1.5 V)                   // V DUT expected, avoid glitches
    .min(1.25 V).max(1.75 V)            // measurement limits
    .iForceRange(25 mA).iForce(10 mA);  // the force current
ifvm.execMode(TM::PVAL).execute();
```

Please note that the force voltage is defined via the “`clamp()`” parameter. For a pure force scenario the `SPMU_TASK` execution mode has to be set to “*FORCE_ON*” instead of “*PVAL*”.

Current force operation is restricted to the actual current range. What needs to be considered is that the DPS128 current ranges do not span from zero to the maximum of the range.

For HC type of channels the current ranges are 20% to 100%, i.e. in the 1 A range you can program current force events in the range from 200 mA to 1 A. In the 200 mA range you can program from 40 mA to 200 mA. This condition also applies for ganged channels, i.e. for the 1 A range the programmable current range is n x 200 mA (min) to n x 1 A (max).

For HV type of channels the current ranges are 10% to 100%.

Operating in force current mode there is a not to be violated dependency between the programmed force voltage and the actual output voltage (clamp voltage). The absolute difference for both voltages must be at least 500mV.

|(Vforce - Vactual)| > 500mV

If violated output stability is not guaranteed. The output current can be much higher than the programmed current limit and oscillations may occur.

Operating in a low impedance scenario (<2 Ω), it is recommended to add a 1 Ω resistor into the force path before the force-sense connection to avoid possible oscillations.
By adding this resistor to the force path the output voltage range of the DPS128 channel gets reduced. The maximum possible output current vs. output voltage by adding the resistor, is depicted on the following V-I curve (HC type of channel).

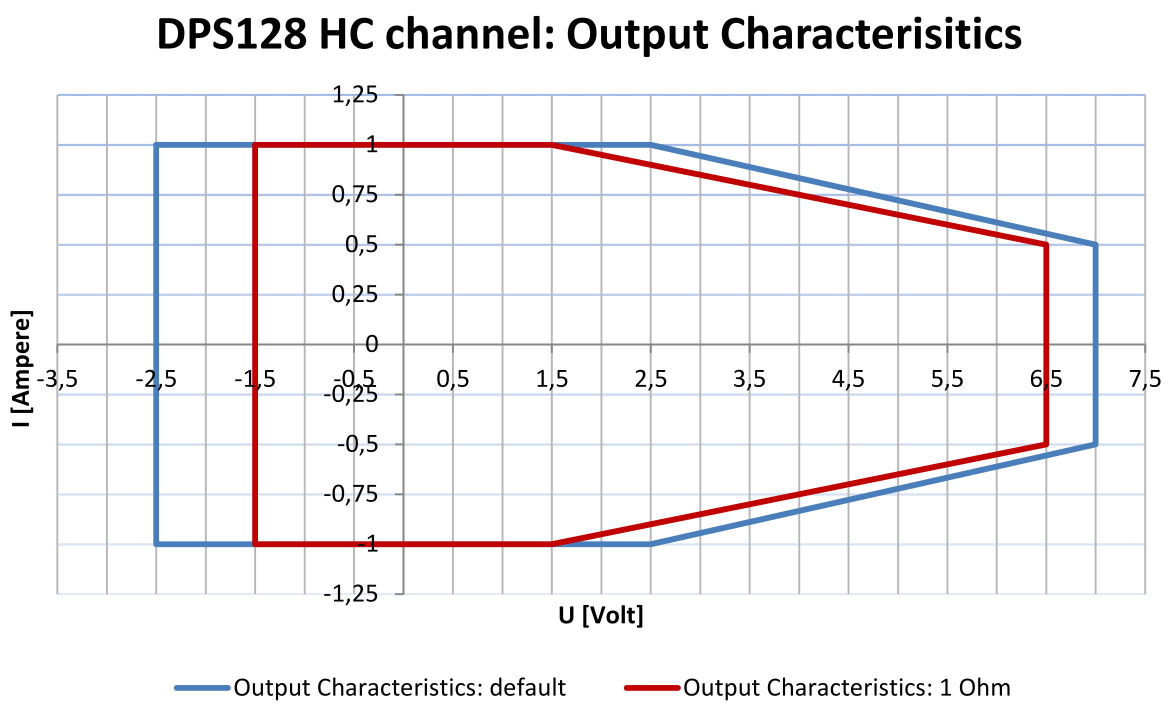


Figure 35: HC type output characteristics with 1 ohm additional resistance in force path

4.3.2 High Accuracy "FI Mode"

DPS128 provides a high accuracy current force mode - "FI Mode" when operating in PMU instrument mode. This feature is not available when operating in DPS instrument mode.

The working model is the same as for normal current force operations, i.e. the clamp circuitry is used. On top of this a digital feedback loop is implemented to improve the native current force accuracy (refer to Specification for more details).

The feedback loop working model can be described as follows:

1. Set specified force current
2. Loop ()
   - 2.1 Measure force current internally
   - 2.2 Correct force current

The correction loop is repeated until the high accuracy specification is met or a maximum of 32 loop iterations are executed. Minimum execution times are documented in chapter "Minimum Event Execution Times in Dynamic Operation".
If correction fails Smartest will force the actual corrected force value, even out of spec.
A flag is available to indicate whether the feedback loop was successful or not. The flag can be accessed using "`DPS_AUX_TASK`" Testmethod API.

"FI Mode" operation is based on the same use model as the standard current force operation. To enable "FI Mode" following Testmethod APIs need to be used:

- "Code-based" static operation: "*SER_HAC*" | "*PAR_HAC*" execution mode parameter of "`SPMU_PIN_TASK`"
- "Pattern-based" dynamic operation: "`EVENT_IC_HAC`"

Alternatively the mode can be programmed directly in the Pattern Editor.
Refer to TDC topic 144527 (`EVENT_IC_HAC`), 131274 (`SPMU_PIN_TASK`) and 141283 (`DPS_AUX_TASK`) for more details.

Please note that forcing into a low impedance load requires some additional resistance in the force path. Refer to chapter "General Use Model" for details.

4.3.3 Voltage Measurement (IFVM)

For DC Scale cards it is possible to measure voltage while in the current force mode ("IFVM"). For this kind of operation DPS128 implements some virtual voltage measurement range. The range defines some not to be violated dependency between the actual voltage at the pin-line and the programmed voltage at the DPS pin.

The voltage measurement will return some "out-of-range" message if following limits are violated:

- DPS128 HC type of channels: (Vactual - Vforce) <= -8 V
- DPS128 HV type of channels: (Vactual - Vforce) <= -5.1 V
  (Vactual - Vforce) >= 8.3 V

Vactual: Actual (measured) voltage at the DPS pin
Vforce: Programmed clamp voltage (force voltage) of the DPS pin

5 Compatibility (DC Scale DPS32/MSDPS)

5.1 Enabling of Dynamic Operation

To enable the dynamic operation it is required to set parameter "test points" in the ".technology" file. This is the same for all DC Sale cards.

Additionally DPS128 requires the 256 waveform timing bundle to guarantee proper operation. This is related to some SmarTest / DPS128 internal working model.

The timing bundle is a global system resource and specified in the model or device license file via keyword "timing_bundle.

"timing_bundle" specifies the number of waveforms, drive edges, and receive edges to be bundled together for all channels.

```
timing_bundle = w32-d8-r6 | w256-d8-r8
```

timing bundle	number of waveforms	number of drive edges	number of receive edges
w32-d8-r6	32	8	6
w256-d8-r8	256	8	8

5.2 Sequencer Rate & Vector Memory in Dynamic Operation

DPS128 implements a single memory controller / single memory per Test-Processor. In other words 8 channels share 1 memory controller and 1 physical memory. This creates restrictions in the available memory per channel, as well as the sequencer run rate.

- Memory per channel limited to 28 M
  - ➔ This is less than DPS32 (56 M), but also less than PS1600 (112 M)
- User period limited to 50 MHz
  - ➔ DPS128 cannot participate in sequencer runs which run faster than 50 MHz, even if DPS is not doing anything. This is also true for primary activation.

The consequence that occurs out of this is that a multiport setup is required, as soon as the DPS works in dynamic mode (testpoints in .technology file > 0) and as soon as one of the below conditions is violated.

- requires more than 28M on any installed channel, also digital
- test items run faster than 50MHz

This is just not about compatibility, but about any general setup in dynamic case. Without Multiport the card with the lowest amount of vector memory defines the overall maximum available. In other words SmartScale (i.e. PS1600) is limited to max 28 M vector memory (default port), as soon as DPS128 is deployed. Introducing Multiport allows utilizing the full available (licensed) memory per card.

Also, original test items running faster than 50 MHz require some special treatment to guarantee synchronous execution between DPS128 channels and non-DPS128 channels. This is about placement of test points at certain vectors / cycles.

Example below illustrates the required test point vs. cycle conversion to keep synchronous execution in respect to the execution time.

Original : DPS32 running @ 160 MHz
Converted : DPS128 running @ 40 MHz

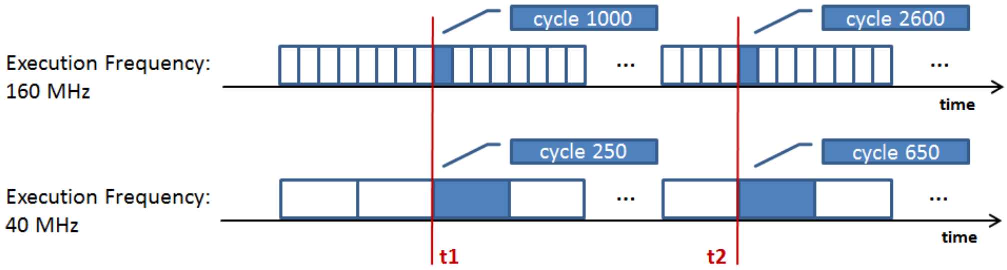


Figure 36: Test point placement compatibility for DPS32 frequencies > 50Mhz

5.3 Current Ranges & Auto-Ranging

DPS128 implements different current ranges than DPS32 and MSDPS. Additionally the force and measure range is combined which requires an auto-ranging approach in the measurement case.

All of the above leads to the following incompatibilities:

- Current force ranges cover 20 % to 100 % (HC type) or 10% to 100% (HV type) of the range, i.e. 1 A range (HC type) is defined as 200 mA – 1000 mA. (DPS32: 1% to 100%)
- Current ranges do not overlap in all places, which might be a problem when the actual flowing current is close to the range limits, i.e. possible constant current situation
- Only certain current force / measure range combinations are allowed by SmarTest. These are so-called virtual combinations, as the hardware can program a common range only (auto-ranging)
- Down-ranging is possible to any supported force / measure range combination
- Up-ranging is only possible to the original range upfront (restore range approach)

Additional incompatibilities in dynamic "pattern based" operation:

- Auto-ranging adds settling times that need to be considered within the pattern. Based on the final measure range a certain amount of pad-vectors or sequencer instruction "*Wait*" has to be inserted to fulfill the required range change time
- Auto-ranging is implemented for set range events (except ganged channels)
- A range restore event is forced after the pattern execution (if a change happened upfront)

Please refer to the chapter "Current Ranges & Current Force/Measure Scenarios" for more details.

5.4 Measurement Range Total Settling Times

Setting a measurement range requires to wait some amount of time until the range has settled in hardware. This is required to achieve the specified accuracy for that range. Additionally a certain amount of wait time might be introduced by the applied range change event itself.
The first one is typically referred to as settling time, while the sum of both is referred to as total settling time.

DPS32 and DPS128 implement different measurement ranges with different total settling times. With regard to the DPS128 total settling times are higher, due to basic hardware settling time differences / requirements, but also due to the required auto-ranging algorithm.
Please note that the total settling time is dominated by the hardware range settling time.

In the static “code based” operation total settling times are transparent to the user, i.e. SmarTest takes care of this internally.
In the dynamic “pattern based” operation the user has to consider the total settling time when setting up the test points in the pattern. This is typically done by adding pad vectors or applying the sequencer instruction “Wait” after any range change event.

DPS32 vs. DPS128 range mapping and total settling times:

DPS32			DPS128		
Range	**Total Settling Times [µs]**	**Specified Capacity [µF]**	**Range**	**Total Settling Times* [µs]**	**Specified Capacity [µF]**
100µA	4000	10	125µA 25µA/12.5µA	10020 – 11310 80020 – 87320	2.2 2.2
2mA	1500	10	2.5mA 1.25mA 250µA	1340 – 2020 1340 – 2020 10020 – 11310	4.7 2.2 4.7
50mA	150	10	25mA/12.5mA	520 – 790	15
1.5A	50	No limit	200mA/100mA	520	47
1.5A	50	No limit	1A	0	220
V to I (keep I Range)	50	No limit	V to I (keep I Range)	500	No limit
I to V	50	No limit	I to V	500	No limit

*: For DPS128 the total settling time includes the auto-ranging sequence / requirement and depends on the initial force range. For further details refer to chapter “Settling Times in “Pattern-Based” Operation”.

Note that the total settling time only needs to be considered when changing the range. In other words, performing a sequence of measurements within the same range does not require additional settling time (except the sample & hold time of the measurement itself).

5.5 Minimum Event Execution Times in Dynamic Operation

In “pattern-based” dynamic operation certain minimum execution times per event exist. If the times are violated an error will be thrown after the pattern run. This error is of global nature, i.e. it’s not possible to tell which event within the pattern is responsible for the violation.

- Minimum vector distance between events within a pattern:
  32 for any action (same as DC Scale DPS32)

- Minimum vectors after any event (especially at the end of a pattern):
  32 for any measure action
  8 for all other actions

- Minimum execution times :
  (0.1) µs after any compare event
  (2.7) µs after any force event
  (230 87422) µs after any high accuracy current force event (FI-mode)
  (6.5 + 5*n) µs after any measure event (n = number of samples)
  (8.5 + 5*n) µs after any combined force/measure event (n = number of samples)
  (10 ... 7320) µs after any range change event

Minimum execution times for range change events:
Minimum time for range change events depends on type / combination of selected force and measure ranges and do not consider range and device settling. For more information about range settling times refer to chapter “Settling Times in “Pattern-Based” Operation”.

Mimimum Range Execution Times in µs (pure auto-range execution time, without measure range settling):

From Range	To Range					
	1A*	200mA, 100mA*	25mA, 12.5mA*	2.5mA, 1.25mA*	250µA, 125µA*	25µA, 12.5µA*
1A*	0	20	290	700	1310	7320
200mA, 100mA*	10	0	20	440	1050	7060
25mA, 12.5mA*	10	10	0	20	640	6650
2.5mA, 1.25mA*	20	10	10	0	20	6040
250µA, 125µA*	20	20	10	10	0	20
25µA, 12.5µA*	20	20	20	10	10	0

Range Down | Restore Range

*: DPS128HC only

Figure 377: Minimum execution times matrix of range event

Minimum execution times for high accuracy current force event (FI-mode):
Minimum execution times depend on channel type, the selected current range and the amount of iteration required to make up the required accuracy.

Below table shows the following execution times
- Minimum (N = 1) : a single loop iteration took place
- Average (N = 4) : this is the typical loop iteration amount to make the accuracy
- Maximum (N = 32): all iterations have been executed. Most likely correction failed

FI Correction Times HC type (min - average - max)

Range	Execution Time / µs		
	N = 1	N = 4	N = 32
1.0 A	238,5	564	3602
200 mA	278,5	664	4262
100 mA	518,5	1264	8222
25 mA	518,5	1264	8222
12.5 mA	1478,5	3664	24062
2.5 mA	1478,5	3664	24062
1.25 mA	1478,5	3664	24062
250 µA	1478,5	3664	24062
125 µA	1478,5	3664	24062
25 µA	1478,5	3664	24062
12.5 µA	5318,5	13264	87422

FI Correction Times HV type (min - average - max)

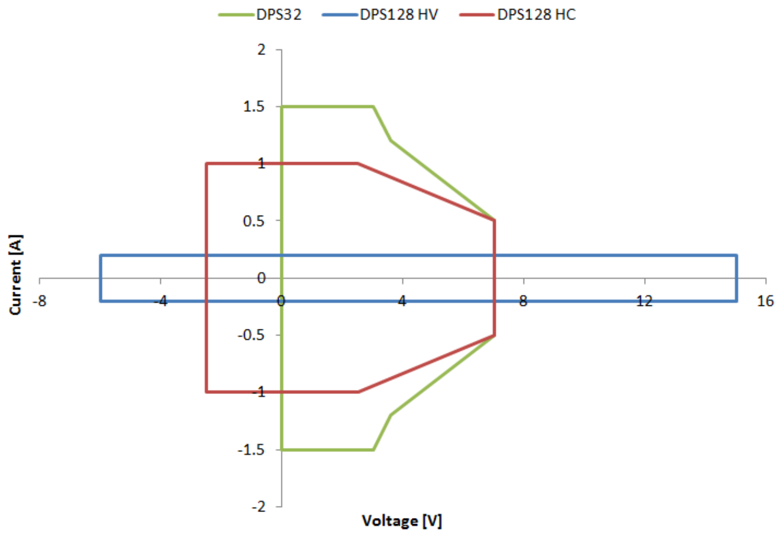
Range	Execution Time / µs		
	N = 1	N = 4	N = 32
200 mA	278,5	664	4262
25 mA	518,5	1264	8222
2.5 mA	518,5	1264	8222
250 µA	2758,5	6864	45182
25 µA	5318,5	13264	87422

Figure 388: Execution times of FI-mode force events

5.6 Output Characteristics and Overall Current Budget

DPS128 provides 4 times the channels than that of DPS32, but delivers less than 2 times the overall current budget. The situation is similar for MSPDS.

	Channels	Voltage Range	Current per Channel	Current per Board (source / sink)
DPS128 HC	128	-2.5 V .. +7 V	1 A @ 2.5 V	66 A / 58 A @ 2.5 V
DPS64 HC	64	-2.5 V .. +7 V	1 A @ 2.5 V	64 A / 64 A @ 2.5 V
DPS128 HC/HV	64	-2.5 V .. +7 V	1 A @ 2.5 V	62 A / 54 A @ 2.5 V (only HC used)
	64	-6 V .. +15 V	0.2 A	12.8 A / 12.8 A (only HV used)
DPS128 HV	128	-6 V .. +15 V	0.2 A	13.2 A / 24.4 A
DPS32	32	0 V .. +7 V	1.5 A @ 3 V	48 A @ 3 V
MSDPS (4 channel mode)	4	-8 V .. +8 V	8 A	16 A / 8 A
MSDPS (8 channel mode)	8	-8 V .. +8 V	4 A	16 A / 8 A



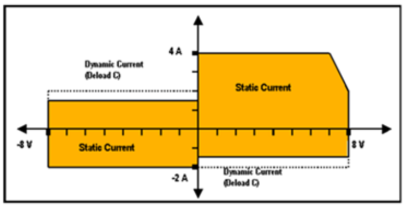


Figure 1: 8-ch mode power diagram

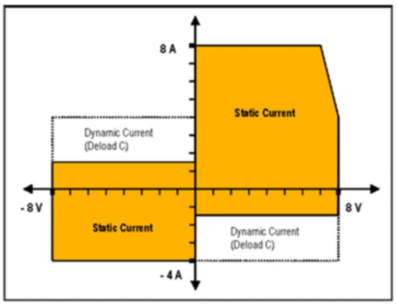


Figure 2: 4-ch mode power diagram

Figure 39: V/I characteristics of V93000 DPS variants

The DPS128 board is supplied by two power domains with each providing half of the overall power available. This implies that the maximum power can only be supplied if the requesting channels are distributed equally over the 2 power domains.
This might lead to situations where a DPS32 cannot be replaced by a DPS128, due to the power vs. channel distribution situation of a DPS128.

For situations where the current budget is critical DPS64 needs to be used together with a Pogo Cable derivate E8023PD (DPS32) or E8023PF (MSDPS).

Please also refer to chapter “Power Domains and Pogo Assignment”.

5.7 Force & Measure Accuracy

Voltage Force	Range	Resolution	Accuracy
DPS128 HC type	-2.5 V .. +7 V	200 µV	±3 mV
DPS128 HV type	-6 V .. +15 V	500 µV	±5 mV
DPS32	0 V .. +7 V	200 µV	±3 mV (0 V .. 5.7 V)
MSDPS (4 channel)	±8 V	300 µV	±4 mV ±0.1% of setting
MSDPS (8 channel)	±8 V	300 µV	±2 mV ±0.1% of setting

Voltage Measure	Range	Resolution	Accuracy
DPS128 HC type	-2.5 V .. +7 V	200 µV	±2 mV ±1 mV (≤ 200 mA force current)
DPS128 HV type	-6 V .. +15 V	100 µV	±4 mV
DPS32	0 V .. +7 V	200 µV	±2 mV
MSDPS	±8 V	1 mV	±2 mV ± 0.1% of reading

Current Force	Range	Resolution	Accuracy
DPS128 HC type	0.2 A .. 1 A 2.5 µA .. 12.5 µA	50 µA 0.5 nA	-3% .. +5% of range -3% .. +5% of range
DPS128 HC type (high accuracy mode)	0.2 A .. 1 A 2.5 µA .. 12.5 µA	50 µA 0.5 nA	± (3 mA ± 0.2% of setting) ± (120nA ± 0.2% of setting)
DPS128 HV type	20 mA .. 200 mA 2.5 µA .. 25 µA	10 µA 1 nA	-3% .. +5% of range -3% .. +5% of range
DPS128 HV type (high accuracy mode)	20 mA .. 200 mA 2.5 µA .. 25 µA	10 µA 1 nA	± (1 mA ± 0.2% of setting) ± (100µA ± 0.2% of setting)
DPS32	1.5 A 100 µA	100 µA 5 nA	± (7 mA + 0.2 % of setting) ± (400 nA + 0.2 % of setting)
MSDPS (4 channel)	8 A, -4 A	1 mA	±20 mA ± 0.5% of setting
MSDPS (8 channel)	4 A, -2 A	1 mA	±20 mA ± 0.5% of setting

Current Measure	Range	Resolution	Accuracy
DPS128 HC type	± 1.0 A ± 12.5 µA	50 µA 0.5 nA	± (1 mA + 0.1 % of reading) ± (50 nA + 0.2 % of reading)
DPS128 HV type	± 200 mA ± 25 µA	10 µA 1 nA	± (300 µA + 0.1 % of reading) ± (60 nA + 0.1 % of reading)
DPS32	±1.5 A ±100 µA	100 µA 5 nA	±(1.5 mA + 0.1 % of reading) ±(100 nA + 0.1 % of reading)
MSDPS (4 channel)	4 A, -2 A (@ -8 V) ±100 µA	150 µA 5 nA	±20 mA ± 0.1% of reading ±100 nA ± 0.1% of reading
MSDPS (8 channel)	4 A, -2 A (@ -8 V) ±10 µA	150 µA 500 pA	±10 mA ± 0.1% of reading ±10 nA ± 0.1% of reading

5.8 External Capacitance Range

DPS128 has slightly different maximum / recommended capacitance ranges than the DPS32 and MSDPS. Despite stability the capacitance has a certain load step response impact.

	DPS128 HC type	DPS128 HV type	DPS32	MSDPS
Min required external capacitance with stability maintained	1 nF .. 10 nF *)	1 nF .. 10 nF *)	1 µF	-
Max supported external capacitance with stability maintained	470 µF/channel 5 mF (ganged)	22 µF/channel 47µF (ganged)	470 µF/channel 1 mF (ganged)	10 µF – 470 µF (depends on current range)
Recommended external capacitance	1 µF.. 100 µF /channel **)	100 nF .. 4.7 µF /channel **)	4.7 µF – 100 µF (depends on current range)	-

*) It is recommended to place a low-ESL capacitance close to the pogo location. The detailed setup depends on the configuration of the DUT interface board. For more details refer to TDC topic 149015.
**) In addition to "Min required external capacitance with stability maintained".

5.9 Profiling

DPS128 implements the same profiling use model as the DPS32. DPS128 has up to twice the sample rate but with a slightly reduced bandwidth.

	DPS32	DPS128
Sample rate	20ksps	50ksps (100ksps)
Bandwidth	25kHz	20kHz

- 50ksps sample rate is available in general for all channels in parallel
- 100ksps sample rate is available for exclusive use of one channel per logical board (channel 1-16, 17-32,33-48,…,113-128)

5.10 AWG & Digitizer Functionality

DPS128 implements the same AWG & Digitizer use model as DPS32. Due to the hardware internals the sampling rate, as well as the available sample memory is reduced.

		DPS32	DPS128
AWG	**Sample Rate**	1Msps	200ksps
	Source Memory	56MB	28MB
	Bandwidth	2.3kHz	5kHz
Digitizer	**Sample Rate**	500ksps	90ksps
	Capture Memory	56MB	28MB
	Bandwidth	25kHz	20kHz

5.11 4-Wire Connection

5.11.1 General Requirements

MSDPS as well as DC Scale DPS channels provide 4-wire connections to compensate for the ohmic losses in the signal path (resistivity, contact resistance) and to compensate for low frequency loading changes. In order to operate the DPS128 channels it is therefore mandatory to connect the force line and ground line to their sense lines as close to the DUT as possible (“remote sensing”).

In case of DPS32 a force to force-sense connection is available within the DPS itself (“local sensing”). Force and force-sense are connected via a so-called *“H_IMP”* relay and some resistance circuit. Note that this local sensing implementation can be considered as a safety feature to avoid any unpredictable DPS states in case there is no force to force-sense connection on the DUT-board. It does not allow for signal path compensation at all, as you miss the complete path up to the DUT (pogo cable, DUT-board).

DPS128 does not implement a local sensing safety circuit, but implements a so-called “force / sense voltage limiter” block. This block guarantees a not to be violated maximum distance between force and force-sense and also acts as a safety feature to prevent unpredictable DPS states.
For standard force operation the “force / sense voltage limiter” works in a way that it will tie up the sense to the force, as soon as the there is a voltage difference greater than 1 V.
When operating in the high impedance measurement mode the limiter block will be shorted and the force sense can float.

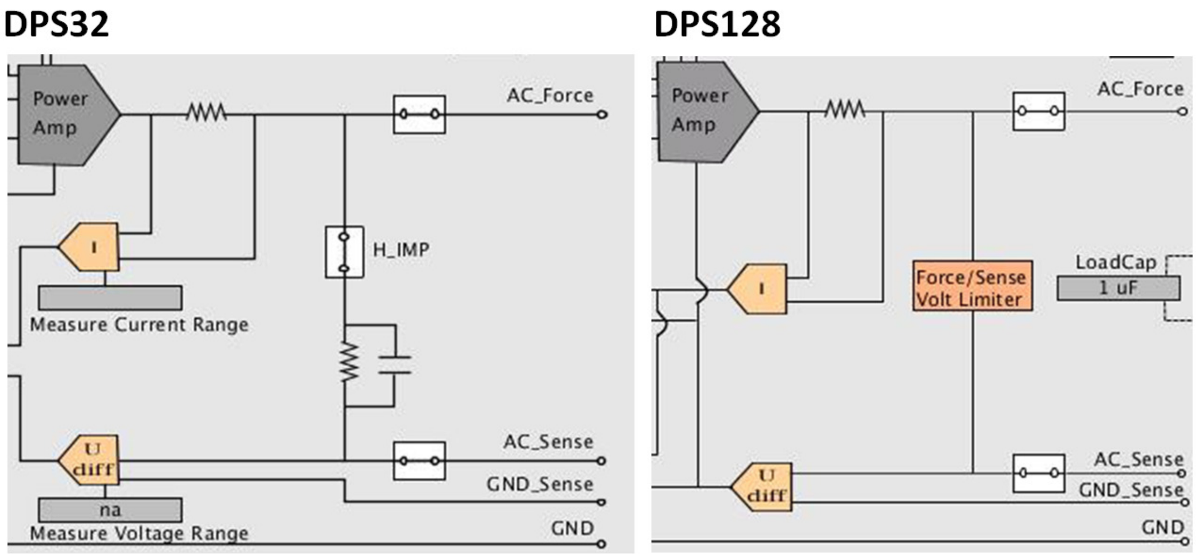


Figure 40: Force / sense line setup in standard operation mode

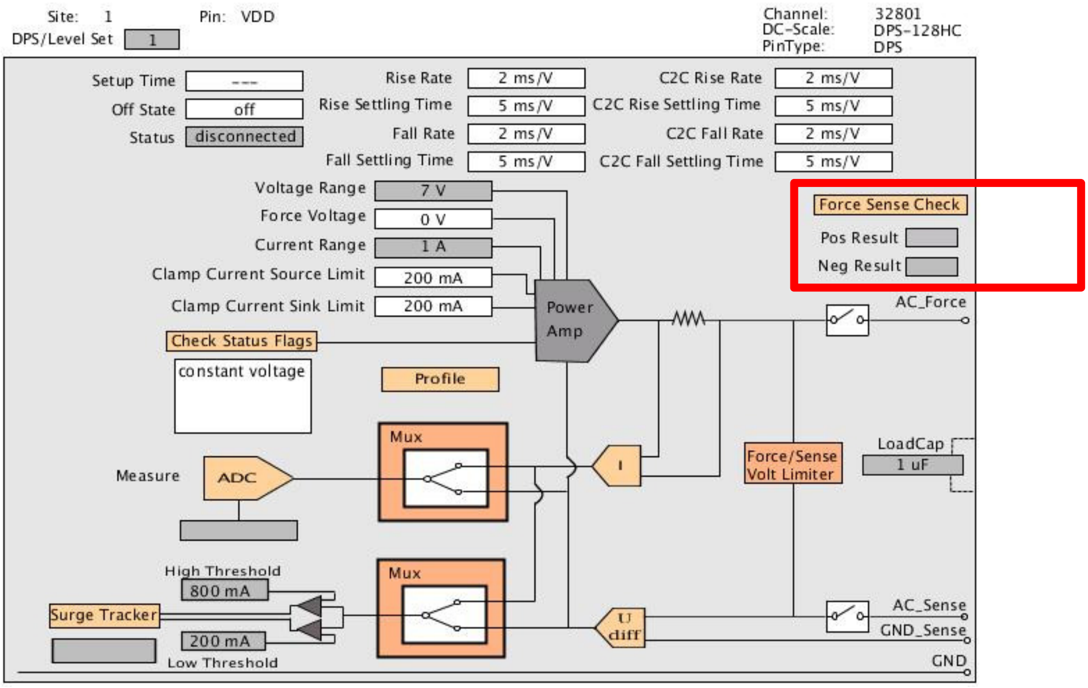
Please note that this safety circuit is only available for the force connection, but not the ground connection. In case the ground to ground-sense connection is missing the DPS128 will operate in a somewhat unpredictable state, i.e. the sense amplifier will float and the DPS will drive some unpredictable force voltage which will result in either a constant voltage or constant current operation mode.

Therefore, and for proper signal path loss compensation, it is mandatory to implement force to force-sense and ground to ground-sense connections on the DUT-board.

Furthermore a general test time hit can be observed if the connections are missing. Smartest polls the status of the DPS while connecting / disconnecting. Without these connections the poll may last until a certain maximum settling time is reached. In case of specifying active HIZ disconnect modes ("`act_min`", "`act_open`"; TDC: 16887) that may take up to 1 second (per disconnect).

Please refer to "DUT Board Design Guidelines" - TDC topic 130656.

The connections can be validated by using the "Force Sense Check" capabilities of the Hardware monitor when in DPS instrument mode.



5.11.2 High Impedance Measurement – Voltmeter

A high impedance measurement can be understood as a (very) low current, voltage type of measurement. To transfer to a low current circuit setup this measurement is done by using a disconnected force line connection. To minimize the leakage current even more the force to force-sense connection is also typically open.

For DPS32 this is accomplished by disconnecting the force line relay as well as the so-called "*H_IMP*" relay that is the connection between force line and force-sense line.

In DPS128 there is no relay, but the "force / sense voltage limiter" block. In high impedance measurement mode this block is shortened and therefore not active. However, this creates some connection between force line and force-sense line that contributes with increased leakage.

As a consequence DPS128 has some increased leakage current of <30 nA (HC type of channel) or <40 nA (HV type of channel), while DPS32 has <2 nA.

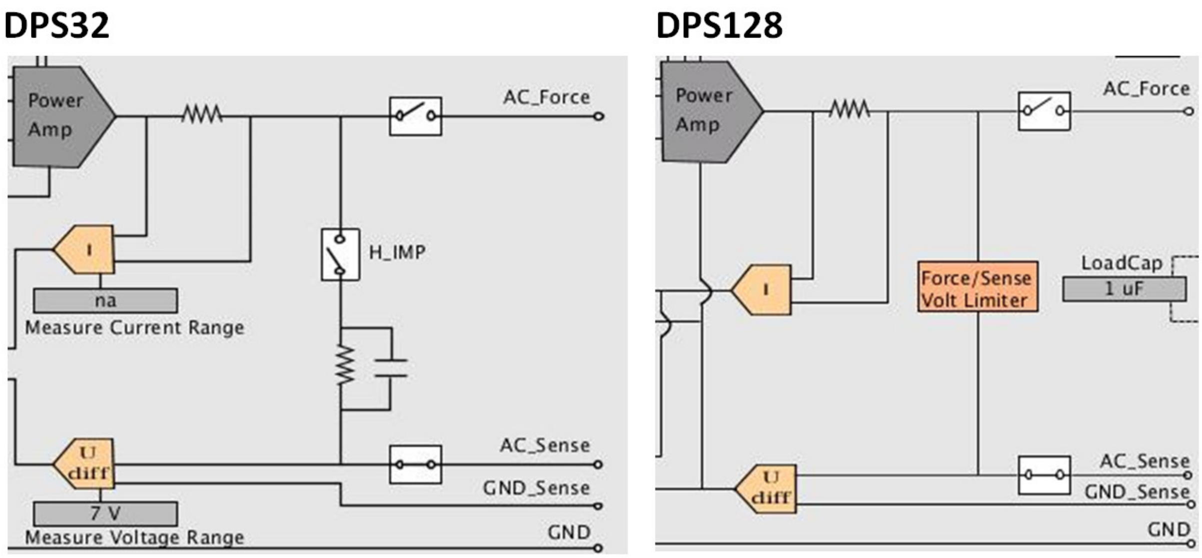


Figure 41: Force/sense line setup for high impedance measurements

5.11.3 Differential High Impedance Measurement – Differential Voltmeter

Differential measurements between the two sense lines allow for floating like measure operations. In this case the force lines (force / ground) are not connected while the two sense lines (force-sense / ground-sense) are hooked up to certain voltage potentials.

For DPS128 this kind of measurement is not supported, as the ground-sense line is not allowed to float, i.e. it needs to be tied to ground.

6 Test Program Migration

DPS128 is a member of the DC Scale family and therefore, compatible to DPS32 in respect to the feature set and basic use model. In the case of MSDPS, DPS128 implements a superset and the same migration steps are required as moving from MSDPS to DPS32.

Even though being feature/use model compatible there are quite some migration steps to take like:

- overall spec ratings (output characteristics)
- trigger based force/measure actions in MSDPS case
- maximum sequencer rate/vector memory limitation in dynamic operation in DPS32 case
- requirements of auto-ranging, especially dynamic case
- power budget (i.e. ganging)
- electrical characteristics (i.e. load step response) and potential adaption of waiting times

The migration effort depends on the base setup. For DPS32 it has to be differentiated between "static" and "dynamic" (test points > 0) test programs, while for MSDPS the setup is considered "static" as long as no triggered force/measure actions have been used. In latter case the MSDPS setup has to be moved to a "dynamic" DPS128 setup.

Major "static" migration tasks:

- overall spec ratings
- power budget (i.e. ganging)
- potential cross-effects of auto-ranging because of its required settling times
- electrical characteristics (i.e. load step response) and potential adaption of waiting times

Major "dynamic" migration tasks:

- all "static" migration tasks
- potential multiport setup based on sequencer run rate and vector memory limitation
- dynamic events; re-arranging of trigger points (auto-ranging)

To keep the migration effort small a special tool-kit is provided.

6.1 Tooling

DC Scale DPS128 migration tool-kit addresses typical "offline" migration topics like configuration, level setups, but also multiport timing and vector generation in "dynamic" case.

It is not able to handle "online" migrations steps like general port synchronization, DPS port period alignment (pattern test points) and electrical performance related issues. It is also not able to handle testmethod based setups and algorithms (i.e. EVENT API based "dynamic" setups or MSDPS based trigger setups).

The tool-kit is part of SmarTest and can be found at
"/opt/hp93000/soc/prod_pws/migration/DPS128_conversion"

There are individual (offline) tools to support the following migration steps:

Pin Configuration / Pogo Cable Configuration

- converts model file (DPS section and pogo-cabling section)
- converts pin configuration file (incl. gang channels/groups)

Level Specifications

- scans the level setup file to detect incompatibilities with respect to specification (no modification)
- does not support DFTD statements and testmethod based setups

Sequencer Rate: Timing and Vector Setups

- modifies pin configuration: creates/splits ports into dps port and digital port
- scans testflow file for items running >50MHz and converts them
    - modifies the timing file: adds new equation sets
    - modifies the pattern file: converts labels to multiport if required
    - updates the testflow file according to above modifications

For more information please refer to the “DPS128 Migration Tool” documentation or TDC topic 137963.

6.2 Potential Issues and Solution

Listing of known migration issues and their solutions, which are not (or only partially) covered by the Migration Tool-Kit.

Category: Power Budget	**Affected Operation Mode:** Static, Dynamic
What	Current level primary setup exceeds overall current budget
Error Message	ERROR: Power budget of boards 225,226 is exceeded!
Solution	1. Introduce voltage ranges (“vout_frc_rng”) to ease the power situation 2. Alternative: Decrease the current clamp limit in level equations. Utilize the separated source/sink limits, i.e. reducing the sink limit shows higher effect than reducing the source limit

Category: Level	**Affected Operation Mode:** Static, Dynamic
What	Current clamp range does not fit to selected current range
Error Message	EQSP - WARNING: Pin "VDD", DPS Set 2536001: Clamp current limit is smaller than minimum value and will be auto-corrected.
Solution	DPS128 HC type clamp ranges are 20% to 80%. 1. Lower current range in level equations (“iout_clamp_rng”) to make the current limit (“ilimit”) >20% of the range 2. Alternative: Increase current limit in level equations (“ilimit”) to be >20% of the selected current range

Category: Timing	**Affected Operation Mode:** Static, Dynamic
What	Default timing set (fixed timing) of existing device setups define a default period (10ns) that is too fast for DPS128
Error Message	ERROR: PCLK - Period 10 ns is out of range due to HW/License restrictions. Valid range is 20 to 31250 ns for waveform set 1.
Solution	Open timing setup file from console using some editor. Search for FW command “PCLK” and change from “PCLK 1,1,10,(@)” to “PCLK 1,1,20,(@)”

Category: Multiport, Timing	**Affected Operation Mode:** Dynamic
What	Periods of dps128 ports cannot be set to wanted value, but get rounded to fit common clocking requirements. → DPS port gets out-of-sync over pattern run time
Error Message	period of port "port_ dps128" in specification "my_equation", timing set 1 changed to 22.057201ns (wanted period 20ns).
Solution	1. Change dps port period value to be of friendly relationship (integer divider) to the master port 2. Alternative: Use clock per pin feature

Category: Multiport, Patterns, DC_tml	**Affected Operation Mode:** Dynamic
What	Multiport Burst does not support STSV (“stop vector”) sequencer instruction. Typical use case: ProductionIDD, ProductionIDDQ
Error Message	ERROR: SQGB - Cannot set mode STSV for a burst label. Use STSQ with a cycle number instead. ERROR: SQGB - Mode cannot be activated
Solution	Convert from stop vector to stop cycle. Add port name to stop cycle, i.e. 2000@port

Category: Multiport, Patterns, DC_tml	**Affected Operation Mode:** Dynamic
What	dc_tml.DcTest.ProductionIddq requires stop cycle number with port specification
Error Message	ERROR: SQGB - Start label is a Multiport Burst Label - per port value is expected ERROR: SQGB - Mode cannot be activated
Solution	Add port specification to stop cycle number, i.e. 2000@port

Category: Multiport, Patterns	**Affected Operation Mode:** Dynamic
What	Inconsistent number of “ctim” , “clev” instructions within same sequencing group → typically affects newly created dps128 port in case “ctim”,”clev” has be used on other ports
Error Message	ERROR: Number of CTIMs and CLEVs is inconsistent on Ports “port_other” (CLEV/CTIM count 2) and “port_ dps128” (CLEV/CTIM count 0). Sequencers would hang. ERROR: FTST? - CExternalException
Solution	1. Apply different ”Sync Group” for dps port and other ports in pattern editor 2. Alternative: create multiport timing (if not already available) and add dps port to different sequencing group

Category: Testpoints, Settling Times	**Affected Operation Mode:** Dynamic
What	Test points need to be adapted to fit the DPS128 settling times.
Solution	Change the test point anchor in the pattern to fit the settling time requirements. It might be necessary to adapt the pattern also, i.e. add additional pad vectors to make sure enough vectors are available. The final solution depends on the actual use model 1. If events are defined in pattern editor: Change the test point anchor in pattern editor 2. If events are defined by using EVENT TM-API: Change the test point anchor using EVENT TM-API

Category: Testpoints, Current Measurements	**Affected Operation Mode:** Dynamic
What	Current measurements require a restore range event, if measurement is executed in a smaller range than originally setup in level equations
Solution	Add a restore event to the pattern. Make sure you fulfill any settling time requirements. The final solution depends on the actual use model 1. If events are defined in pattern editor: Add the restore event in pattern editor 2. If events are defined by using EVENT TM-API: Add the restore event using EVENT TM-API

Category: Testpoints, Multiport	**Affected Operation Mode:** Dynamic
What	DPS32 is able to run a vector period of 2.5ns while DPS128 minimum period is 20ns In case the original pattern is running a vector period < 20ns the DPS128 channels need to be moved to a separate port. If the original pattern holds testpoints vector synchronization needs to be done.
Solution	In dependency to the master port period (<20ns) and the dps port period (>=20ns) the testpoints in the dps port need to be adapted to keep the original vector relation to the master port. See also “Sequencer Rate & Vector Memory in Dynamic Operation” for more details. . The final solution depends on the actual use model 1. If events are defined in pattern editor: adapt the test points in pattern editor 2. If events are defined by using EVENT TM-API: adapt the test points by using EVENT TM-API

Category: Testpoints , Measurement Samples	**Affected Operation Mode:** Dynamic
What	DPS128 needs some amount of samples to achieve the specified accuracy. The amount of samples depends on the selected measurement current range. The range and the amount of samples is typically different to DPS32. For dynamic operation mode the samples need to be specified manually in the measurement event, while in static mode the required samples are automatically done by FW.
Solution	Check specification for required samples. Adapt the sample amount for the measurement event. The final solution depends on the actual use model 1. If events are defined in pattern editor: adapt the measurement samples in pattern editor 2. If events are defined by using EVENT TM-API: adapt the measurement samples by using EVENT TM-API

7 DPS128 Setup & Programming

7.1 Basic Testprogram Setup

Using DC Scale DPS128 resources it has to be decided whether to operate in the static “code based” or dynamic “pattern based” operation. This decision affects the system as a whole.

Static “code based” operation:

- Sequencers are disabled, i.e. not taking part in any pattern run
- No triggered actions like vBump (DC Scale is sequencer triggered)
- Asynchronous measurements - “code based”

Dynamic “pattern based” operation:

- Sequencers are active, i.e. taking part in any pattern run
- Pattern synchronous actions (force/measure) possible - “pattern based”
- “SmartCalc” support
- Multiport required when running > 50 MHz and/or using more than 28 M vector memory on digital channels

To enable Dynamic operation 2 steps are required.

1. Test points
   Parameter “test points” in .technology file (via “Change Device” Dialog)

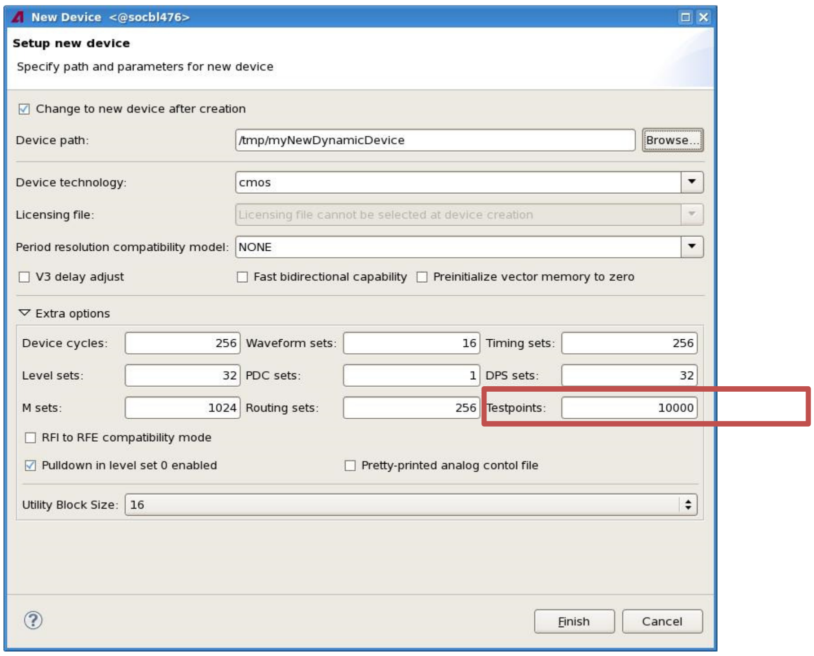


Figure 42: Device technology setup

2. 256 Waveforms

   256 waveform timing bundle specified in model and / or device license file

```
GLOBAL
  testhead = TH_8CC                          # TH_1CC | TH_2CC | TH_4CC | TH_8CC
  dut_interface = SOC_8GROUP                 # SOC_4GROUP | SOC_8GROUP | HSM
  hpib_interface = vxi11/e5810a/192.168.0.100/gpib0
  timing_bundle = w256-d8-r8
#  protocol_aware = ON
#  utility_lines = 64, type=static, pogo_size=16
```

Please refer to the chapter “Enabling of Dynamic Operation” for more details.

7.2 Defining the Basic Instrument Mode

A DC Scale channel can be defined as a conventional DPS Instrument (DPS mode) or a pure DC resource (SIG mode).

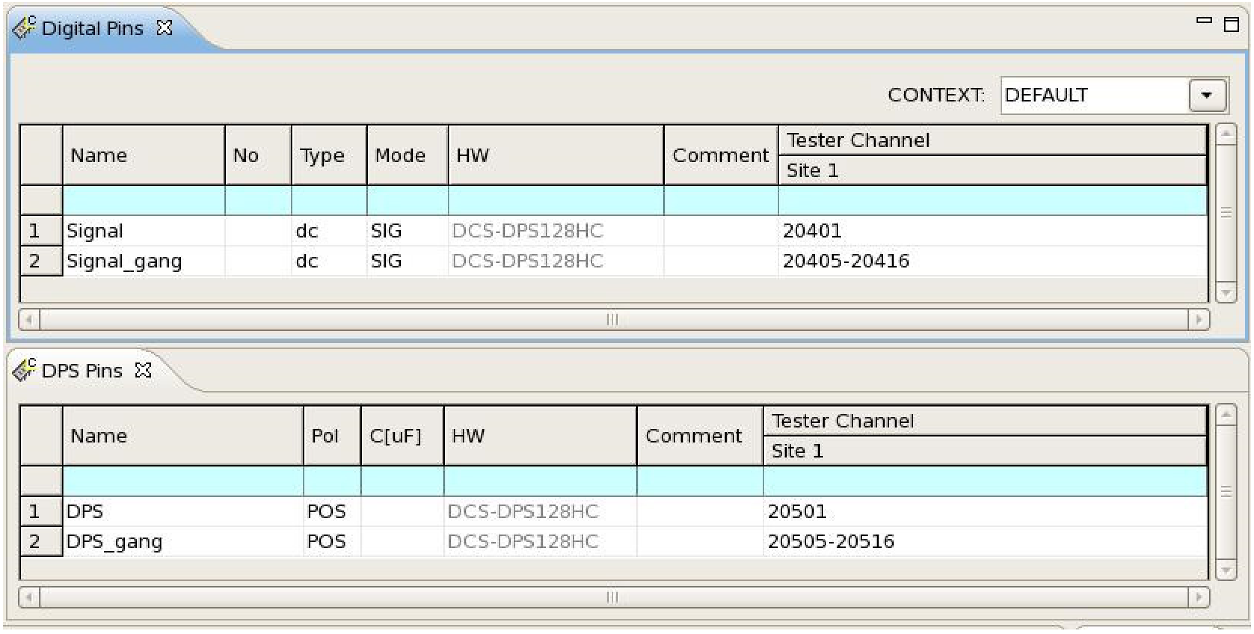


Figure 43: DPS / PMU mode pin setup

To setup in the PMU (SIG) mode use the “Digital Pins” tab of the pin configuration. To setup in the DPS mode use the “DPS Pins” tab of the pin configuration.

The major difference between these two instrument modes is the use model.

Refer to next chapters, explaining about the differences.

7.2.1 DPS Instrument Mode

DC Scale channels defined in DPS mode join the general connect / disconnect sequence. In order to perform accordingly the channels have to be defined in the level setup.

Figure below shows an example of a DPS setup in the equation block of the level file:

```
DPSPINS VDD
vout = 2                    #[V]
vout_frc_rng = 2.4          #[V]
iout_clamp_rng = 1000       #[mA]
ilimit = 600                #[mA]
t_ms = 2                    #[ms]
offcurr = act
#disable_const_curr_check
```

There are some special considerations for certain key-words in DPS128 case:

- *vout_frc_rng*
    - specifies the force voltage range with unit [V]
    - DPS128 has indiviual ranges between 2.25 V and 7 V for HC type; 15 V fix HV type
    - when it is not specified, highest voltage range is selected
- *ilimit | ilimit_sink, ilimit_source*
    - programs the current clamps, either both together (*ilimit*) or individual (*ilimit_sink, ilimit_source*)
    - note that the power diagram puts a constraint here. For example, in the case of DPS128, when vout is set to 7V, according to the power diagram, the maximum loading current allowed is 500mA (HC type).
- *iout_clamp_rng*
    - specifies the current clamp range with unit [mA]
    - if it is not specified, highest current range get selected
- *connect_state*
    - specifies the connection mode of the DPS, i.e. high impedance, un-gang or normal
- *disable_const_curr_check*
    - suppresses warnings in case current hits clamp current range
    - needed in case DPS is used as constant current source

It is very important to keep in mind that using "*disable_const_curr_check*" suppresses not only the warning message but also the checking of DPS voltage settling.

So when doing a normal connect sequence (without disabling constant current check) SmarTest triggers the DPS to program the desired voltage. Then it verifies the voltage until it is stable. That means that SmarTest actually blocks all other actions until all voltages are stable.

When "*disable_const_curr_check*" is set, SmarTest does not do this settling check but only programs the voltage and keeps on going. It means when you specify "*disable_const_curr_check*" on 'normal' setups then the voltage at the start of your functional test might not yet be completely settled and so you might run into correlation issues.
To use DPS as a current source and/or to perform an IFVM task (current force voltage measure), it is recommended to set this key word.

Please refer to TDC topic 16887 for more details on the level equation part.

The main application focus of the DPS is mode is to supply devices with power, but it supports voltage force and measure as well as current force and measure.
To force a current the clamp circuitry is used. For current measurements DPS128 will do auto-ranging. Also refer to chapter “Current Ranges & Current Force/Measure Scenarios”.

For more details on DPS mode operation please refer to TDC topic 140642.

Testmethod programming is done by using following Testmethod APIs:

- Static “code based” operation: `DPS_TASK`
- Dynamic “pattern based” operation: `EVENT + PATTERN_CONTROLLER`

Please refer to the chapter “Testmethod Programming” for more details.

7.2.2 PMU Instrument Mode

In the PMU mode operation DC Scale channels behave similar to a PMU. Therefore, the channels do not join the general connect and disconnect sequence. However, they can be added to the user defined connect and disconnect sequences.

For the PMU operation no equation based setup is required. Therefore all force and measure parameters have to be well defined using the corresponding Testmethod API.

- Static “code based” operation: `SPMU_TASK`
- Dynamic “pattern based” operation: `EVENT + PATTERN_CONTROLLER`

As channels do not participate in connect / disconnect by default it is required to connect the resource before doing any measurement. Despite switching relays the connect sequence is also responsible to setup the correct voltage and current ranges. For the dynamic operation it is required to connect and setup the resource upfront in a static “code based” way.

Voltage force connect:

```
// voltage force connect: force 2.5 V in 3 V voltage range
// 200 mA current range with clamp at 100 mA sink / 150 mA source
SPMU_TASK connect;
connect.pin(sigPin).mode("VFIM").relay("NT")
        .vForceRange(3 V).vForce(2.5 V)
        .iForceRange(200 mA)
        .min(0 mA).max(1 mA)     // dummy, as no measurement
        .clamp(100 mA, 150 mA); // for different sink/source limits
connect.execMode("FORCE_ON").execute();
```

Note:
In case the “`clamp()`” parameters are not specified the “`max()`” parameter will define a common (sink/source) current clamp limit.
It is possible to connect in “*ungang*” mode, i.e. limiting the amount of connected channels to the maximum required. Refer to chapter “Partial Ganging” for more details.
For current ranges < 100 mA the voltage force range is automatically programmed to 7V.

Current force connect:

Current force is implemented via the clamp circuitry that is operating in the constant current source mode. Therefore the direction of the current flow is decided by the potential difference between device output voltage and the programmed clamp voltage (force voltage).
Please refer to the chapter “Current Force Scenario” for more details.

The Testmethod API offers 2 alternatives to implement current force.

Alternative 1 – using a “VFIM” style setup:

```
// current force connect - Alt1:
// force 50 mA into device, 100 mA current range
// 2.75 V clamp (force) voltage in 3 V voltage range
SPMU_TASK connect;
connect.pin(sigPin).mode("VFIM").relay("NT")
       .vForceRange(3 V).vForce(2.75 V)    // the clamp voltage
       .preCharge(1 V)                     // V DUT expected, avoid glitches
       .iForceRange(100 mA).clamp(50 mA)   // the force current
       .min(0 mA).max(1 mA);               // dummy, as no measurement
connect.execMode("FORCE_ON").execute();
```

Alternative 2 – using a “IFVM” style setup

```
// current force connect - Alt2:
// force 50 mA into device, 100 mA current range
// 2.75 V clamp (force) voltage in 3 V voltage range
SPMU_TASK connect;
connect.pin(sigPin).mode("IFVM").relay("NT")
       .vForceRange(3 V).clamp(2.75 V)       // the clamp voltage
       .preCharge(1 V)                       // V DUT expected, avoid glitches
       .min(0 V).max(1 V)                    // dummy, as no measurement
       .iForceRange(100 mA).iForce(50 mA);   // the force current
connect.execMode("FORCE_ON").execute();
```

Alternative 1 also allows connecting in “*ungang*” mode, i.e. limiting the amount of connected channels to the maximum required. Refer to chapter “Partial Ganging” for more details.

Note:
For current ranges < 100 mA the voltage force range is automatically programmed to 7V.

General disconnect:

```
//disconnect: will not adapt the previous settings
SPMU_TASK disconnect;
disconnect.pin(sigPin).relay("NT").disconnect();
disconnect.execMode("FORCE_OFF").execute();
```

Any “`FORCE_OFF`” action will just open the relays without changing the specified current / voltage values and ranges. Take care for hot-switching.

Actual settings can easily be observed by utilizing the Hardware Monitor of SmarTest.

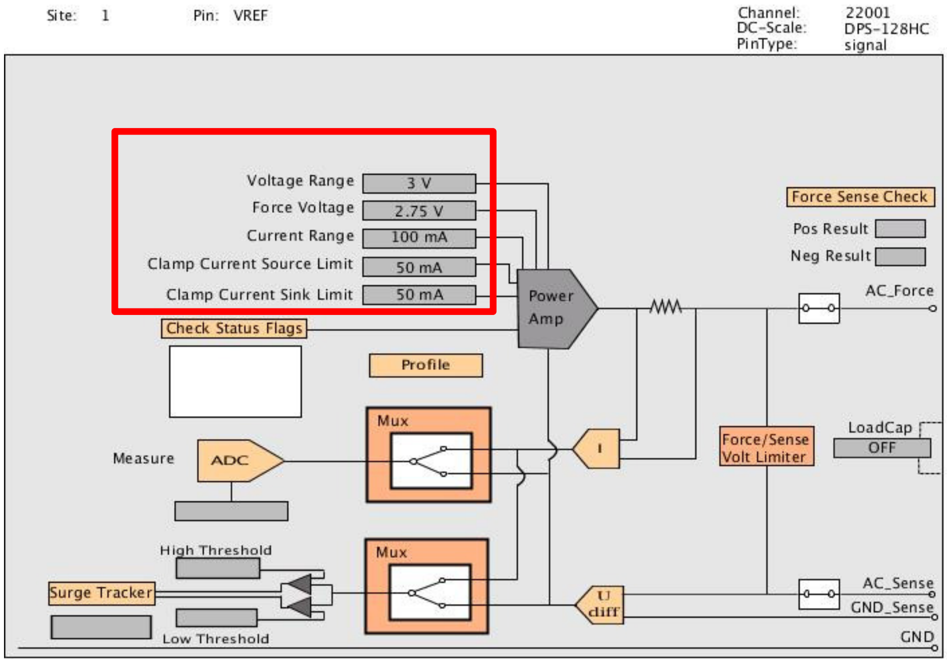


Figure 44: PMU instrument mode IFVM settings in Hardware Monitor

Similar to the DPS mode it is possible to perform a voltage and current force action, as well as voltage and current measurement actions. This also includes un-ganged current measurements.

For more details on PMU instrument mode operation please refer to TDC topic 140645.

7.3 Dynamic “Pattern Based” Operation - Multiport

Both basic operation modes, DPS as well as SIG, require a multiport setup if certain conditions are true.

- vector period > 50MHZ
- pattern memory requirement > 28M

For the first condition a pure timing port is enough, while the second condition requires a vector port.

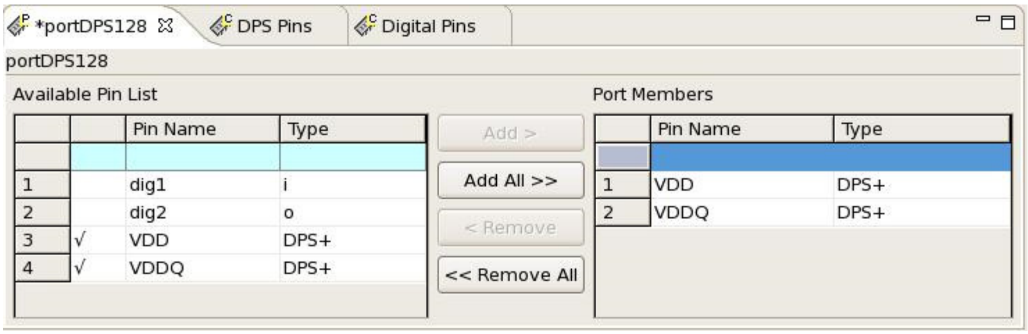


Figure 45: DC Scale multiport setup in pin configuration

DC Scale channels can be defined in a port as digital pins. DC Scale pins can reside in a port with digital pins or by themselves only.
For the timing setup, DC Scale channels only define the vector period.

```
EQNSET 99 "DPS128"
DEFINES portDPS128

SPECS
Frequency [MHZ]  # must be less than 50MHz

EQUATIONS
cycleTime =1000/Frequency

TIMINGSET 1 "default"
period=cycleTime
```

For the vector setup (vector port) the user has to setup multiport burst labels. However, in the case where no dynamic action is required, no special label has to be prepared for the DPS port.

Signal	DC-Scale (Instructions)	Digital (Instructions)
Call# | Grp	DEFAULT	DEFAULT
0	BEND	CALL Functional1
1		CALL Functional2
2		CALL Functional3
3		BEND

Figure 46: Multiport Burst-Label with empty labels for DC Scale port

7.4 Testmethod Programming

The DPS128 supports the static and dynamic operation for both of the DPS and PMU modes. Static programming utilizes different APIs, while in dynamic case the use- and programming model is the same.
In general the use model and Testmethod APIs used are compatible to DC Scale DPS32. New DC Scale DPS128 features are added by either enhancing the existing APIs or adding complete new ones, i.e. “`DPS_AUX_TASK`” to control the comparators.

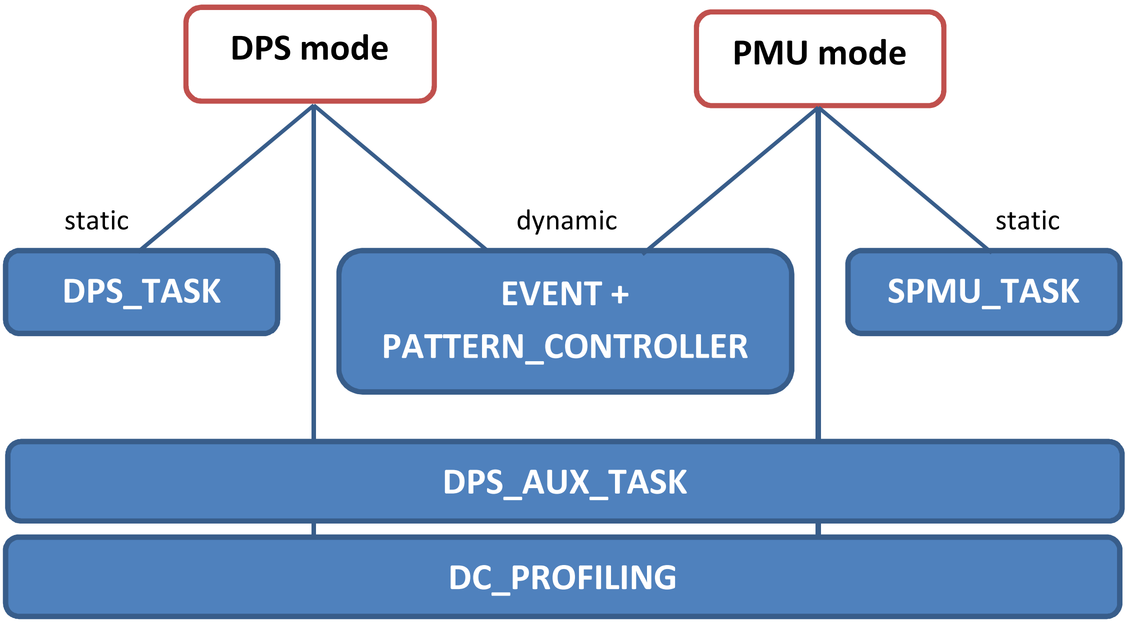


Figure 47: Testmethod API Concept

In general a Testmethod sample code (“static”) can be generated by utilizing the HW Monitor, i.e. doing an interactive voltage measurement will generate a sample code as shown below.

For further details on “Sample Code View” please refer to the TDC topic 108172.

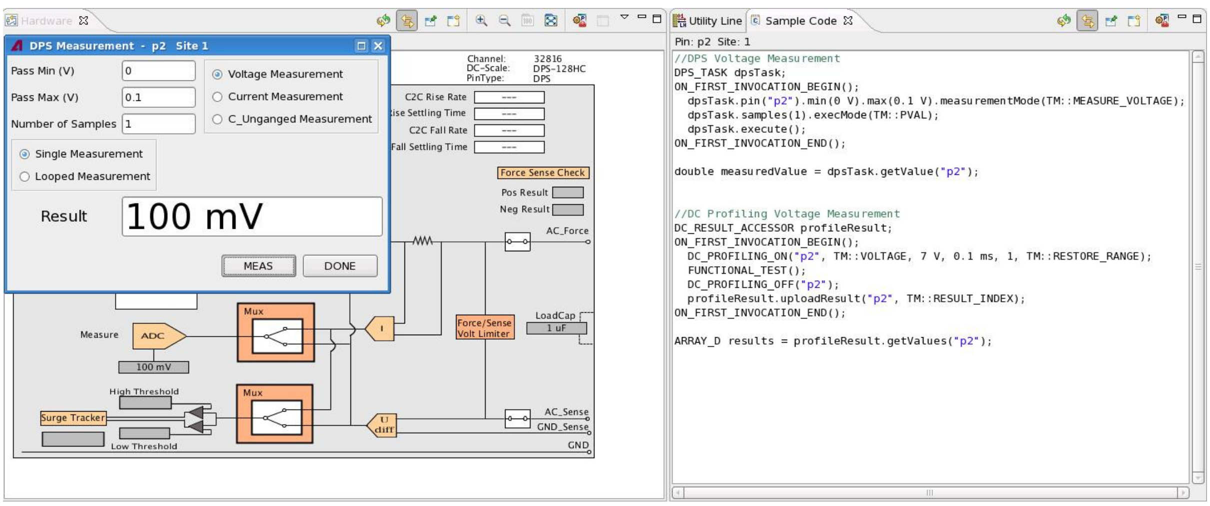


Figure 48: Sample code generation in HW Monitor

7.4.1 DPS Mode Current Measurement

This is the typical Production IDD/ IDDQ use case. For IDDQ the DPS128 use model is different compared to DPS32, as an auto-range/restore range is required.
In “static” operation this is transparent to DPS32 as the ranging will be done internally by SmarTest.
For “dynamic” operation special care has to be taken about the placement of the auto-range and restore-range event.

Please refer to the chapter “Current Ranges & Current Force/Measure Scenarios” for more details on auto-ranging.

7.4.1.1 “Static” Operation

```
//IDDQ current measurement
DPS_TASK measTask;
DPS_AUX_TASK auxTask;

ON_FIRST_INVOCATION_BEGIN();
    measTask.pin("dps128_pin")
            .min(0.001 A)
            .max(0.01 A)
            .measurementMode(TM::MEASURE_CURRENT);

    CONNECT();
    measTask.execMode(TM::PVAL).execute();

    auxTask.pin(“dps128_pin”).checkSetRangeError();
ON_FIRST_INVOCATION_END();

// check for successful ranging
TM::DPS_AUX_TASK_STATUS status;
status = auxTask. getRangeError(“dps128_pin”);
if(status == TM::AUX_TASK_STATUS_FAIL){
   cout << “Warning: Auto-range sequence didn’t succeed!” << endl;
}

double measuredValue = dpsTask.getValue("dps128_pin");
```

For a description of the `DPS_TASK` API refer to TDC topic 55630.

For a description of the `DPS_AUX_TASK` API refer to TDC topic 141283.

7.4.1.2 “Dynamic” Operation

```
//IDDQ current measurement
PATTERN_CONTROLLER measureCtrl;
DC_RESULT_ACCESSOR resultAccess;
DPS_AUX_TASK auxTask;

ON_FIRST_INVOCATION_BEGIN();
      EVENT_RANGE_IM rangeEvent("range");
      rangeEvent.iRange(20 mA, 50 mA);

      EVENT_IM measureEvent("measure");
      measureEvent.limitLow(25 mA).limitHigh(45 mA).averages(4);

      // optional
      // EVENT_RESTORE_RANGE restoreEvent("restore");

      // put events to the pattern
      measureCtrl.label("DC Scale_label","dps128_port")
                  .vecEvents("dps128_pin")
                  .clear()
                  // make sure actual current flow fits the range
                  .insert(10000, rangeEvent)
                  // make sure to have ranging finished
                  .insert(25000, measureEvent)
                  // optional; make sure measurement has finished
                  //.insert(32000, restoreEvent)
                  .setup();

      CONNECT();
      FUNCTIONAL_TEST();

      auxTask.pin(“dps128_pin”).checkSetRangeError();
      resultAccess.uploadResult("dps128_pin",
                           TM::RESULT_RANGE(TM::RESULT_INDEX,0,1));
ON_FIRST_INVOCATION_END();

// check for successful ranging
TM::DPS_AUX_TASK_STATUS status;
status = auxTask. getRangeError(“dps128_pin”);
if(status == TM::AUX_TASK_STATUS_FAIL){
   cout << “Warning: Auto-range sequence didn’t succeed!” << endl;
}

// do some post-processing on values
ARRAY_D results = resultAccess.getValues("dps128_pin");
for(int r=0; r<results.size(); ++r){
      ...
}
```

For a description of the used APIs refer to TDC topics 116911 (`EVENT`), 71357 (`PATTERN_CONTROLLER`), 116944 (`VEC_EVENTS` as part of `PATTERN_CONTROLLER`), 99071 (`DC_RESULT_ACCESSOR`) and 141283 (`DPS_AUX_TASK`).

7.4.2 Dynamic Current Ramp

Dynamic actions use the same programming interface for both operational modes. They are only allowed within a single force current range, as switching the range requires a disconnect sequence.

7.4.2.1 DPS Mode

In DPS mode the actual voltage and current ranges are defined by the level primary setup.

```
// current ramp
PATTERN_CONTROLLER rampCtrl;

ON_FIRST_INVOCATION_BEGIN();
    // abstract ramp definition
    ANALOG_WAVEFORM currentRamp("currentRamp");
    currentRamp.definition(TM::RAMP)
               .min(450 mA)
               .max(550 mA)
               .direction(TM::POS)
               .samples(100);

    // DPS adaptation
    EVENT_WAVEFORM_IC rampUpDps("rampUpDps");
    rampUpDps.iClampLow(currentRamp)
             .sampleVectorDistance(256);

    // put event to the pattern
    rampCtrl.label("DC Scale_label","dps128_port")
            .vecEvents("dps128_pin")
            .clear()
            .insert(31, rampUpDps)   // 1st vector position of wave
            .setup();

    CONNECT();
    FUNCTIONAL_TEST();
ON_FIRST_INVOCATION_END();
```

For a description of the used APIs refer to TDC topics 116911 (`EVENT`), 71357 (`PATTERN_CONTROLLER`), 116944 (`VEC_EVENTS` as part of `PATTERN_CONTROLLER`) and 103395 (`ANALOG_WAVEFORM`)

7.4.2.2 PMU Mode

PMU mode requires the need to connect the DPS channels upfront, as the PMU mode does not take part in the standard connecting sequence. The connect sequence requires the need also to setup the voltage and current range settings.

```
// current ramp
PATTERN_CONTROLLER rampCtrl;

ON_FIRST_INVOCATION_BEGIN();
    // static implementation of connect sequence
    // force 450 mA, 1000 mA current range, 3 V clamp voltage
    SPMU_TASK connect;
    connect.pin(viMeasPins).mode("IFVM").relay("NT")
           .vForceRange(3 V).clamp(3 V)
           .precharge(1 V)      // V DUT expected, avoid glitches
           .min(0 V).max(1 V)   // dummy, as pure connect only
           .iForceRange(1000 mA).iForce(450 mA);
    connect.execMode("FORCE_ON");

    SPMU_TASK disconnect;
    disconnect.pin(viMeasPins).relay("NT").disconnect();
    disconnect.execMode("FORCE_OFF");

    // abstract ramp definition
    ANALOG_WAVEFORM currentRamp("currentRamp");
    currentRamp.definition(TM::RAMP)
               .min(450 mA)
               .max(550 mA)
               .direction(TM::POS)
               .samples(100);

    // DPS adaptation
    EVENT_WAVEFORM_IC rampUpDps("rampUpDps");
    rampUpDps.iClampLow(currentRamp).sampleVectorDistance(256);

    // put event to the pattern
    rampCtrl.label("DC Scale_label","dps128_port")
            .vecEvents("dps128_pin")
            .clear()
            .insert(31, rampUpDps)    // 1st vector position of wave
            .setup();

    CONNECT();
    connect.execute();      // PMU instrument connect
    FUNCTIONAL_TEST();
    disconnect.execute();   // PMU instrument disconnect

ON_FIRST_INVOCATION_END();
```

For a description of the used APIs refer to TDC topics 116911 (`EVENT`), 71357 (`PATTERN_CONTROLLER`), 116944 (`VEC_EVENTS` as part of `PATTERN_CONTOLLER`), 103395 (`ANALOG_WAVEFORM`) and 55670 (`SPMU_TASK`)

7.4.3 Comparator Programming

"DPS_AUX_TASK" is introduced to control the DPS128 comparator thresholds and operation modes. This API is valid for "code-based" static as well as "pattern-based" dynamic operation (TDC topic 141283).

"EVENT_CMP" is introduced to perform compare actions for "match-loop" like use cases. This is supported in "pattern-based" dynamic operation only (TDC topic 142143).

7.4.3.1 Surge Tracker

The explicit activation / de-activation of the tracker imply the use model of having 2 testsuites to control it. There is one testsuite to enable the tracker mechanism; another one to disable and check the status. This allows monitoring over a wide test sequence.

Alternatively it is also possible to restrict the tracker to a single testsuite by activating / de-activating it within the testsuite related Testmethod.

Activation

```
DPS_AUX_TASK auxTask;
ON_FIRST_INVOCATION_BEGIN();
    auxTask.pin("dps128_pin")
           .setMode(TM::SURGE_TRACKER)
           .selectVSensorSetThresholds(-1.0 V, 2.5 V);
    auxTask.execute();
ON_FIRST_INVOCATION_END();
```

Check and Deactivate

```
DPS_AUX_TASK auxTask;
ON_FIRST_INVOCATION_BEGIN();
    auxTask.pin("dps128_pin").checkSurgeTracker(TM::AGAINST_BOTH);

    //turn off
    auxTask.pin("dps128_pin").setMode(TM::DPS_AUX_OFF);
    auxTask.execute();
ON_FIRST_INVOCATION_END();

// Get the result of surge tracker
TM::SURGE_TRACKER_STATUS status;
status = auxTask.getSurgeTrackerStatus("dps128_pin");

switch (status){
  case TM::SURGE_TRACKER_NONE_DETECTED: /* do something */ break;
  case TM::SURGE_TRACKER_LOW_DETECTED:  /* do something */ break;
  case TM::SURGE_TRACKER_HIGH_DETECTED: /* do something */ break;
  case TM::SURGE_TRACKER_BOTH_DETECTED: /* do something */ break;
  default:                             /* do something */ break;
}
```


7.4.3.2 Match-Loop

Match-Loop programming is divided into 2 parts. The first part is the basic compare mode setup which is handled within the testmethod context utilizing `DPS_AUX_TASK` and (optionally) `EVENT_CMP` API.

- `DPS_AUX_TASK` is responsible to active the compare mode, setup the type (current or voltage) and the threshold value (high threshold)
- `EVENT_CMP` is responsible to setup the single threshold compare mode; either low or high

The second part is the match sequence setup which is done in pattern editor utilizing the match-loop related sequencer instructions and, if not done within testmethod context, the setup of the compare event.

Below example will pass the match loop if the retrieved voltage is < 1 V.

```
// "Voltage Compare" - match
DPS_AUX_TASK auxTask;
PATTERN_CONTROLLER compareCtrl;

ON_FIRST_INVOCATION_BEGIN();
    //define compare event and place inside pattern
    EVENT_CMP compareEvent("compare");
    compareEvent.low();

    compareCtrl.label("DC Scale_label","dps128_port")
            .vecEvents("dps128_pin")
            .clear()
             .insert(1000, compareEvent) //test point, test data
            .setup();

    // enable DPS comparator
    auxTask.pin("dps128_pin")
           .setMode(TM::CMP_HIGH)
           .selectVSensorThresholds(0 V/*not used*/, 1 V);
    auxTask.execute();

    CONNECT();
    FUNCTIONAL_TEST();

    //disable DPS comparator
    auxTask.pin("dps128_pin").setMode(TM::DPS_AUX_OFF);
    auxTask.execute();
ON_FIRST_INVOCATION_END();
```


7.4.4 Profiling

Profiling allows the ability to measure a voltage or current during a functional test. It can either been done interactively via the HW-Monitor or in a programmatic way using TM API.

```
//DC Profiling Current Measurement
DC_RESULT_ACCESSOR profileResult;

ON_FIRST_INVOCATION_BEGIN();
    DC_PROFILING_ON("dps128_pin",
                    TM::CURRENT,
                    0.8 A,
                    0.02 ms,
                    1,
                    TM::RESTORE_RANGE);

    CONNECT();
    FUNCTIONAL_TEST();

    DC_PROFILING_OFF("dps128_pin");
    profileResult.uploadResult("dps128_pin", TM::RESULT_INDEX);
ON_FIRST_INVOCATION_END();

ARRAY_D results = profileResult.getValues("dps128_pin");
```

The result can be uploaded to Signal Analyzer using `PUT_DEBUG` API.

For a description of the Profiling API refer to TDC topic 125003.

8 DUT Board Design Guidelines

8.1 Mandatory Force / Sense Connections

The force, force-sense, ground and ground-sense lines of the DPS128/DPS64 have specific connection requirements.

In order to operate the DPS128, you must connect both the force line to the force-sense line and the ground to the ground-sense line. Make the connection as close as possible to the DUT.

For the routing of sense lines follow these rules:

- Route sense lines as traces.
- Connect sense lines to force as close to the DUT as possible.
    - Connect sense lines at DUT supply pins, or
    - Connect them at the closest bypass capacitor
- Treat sense lines as critical signal paths.
    - Separate them from other traces to avoid feeding noise back to the DPS voltage source.

Note:
The DPS128/DPS64 has no local sensing safety circuit (unlike the DPS32). Instead it uses only a force/force-sense voltage limiter block, which acts as a limited safety feature.
If the force and sense connections are missing, several unpredictable DPS states may occur. The same especially applies to missing ground to ground-sense connections.

For more details refer to chapter “4-Wire Connection”

8.2 Power Domains

The DPS128/DPS64 card has two power domains with a dedicated power budget. Each domain is able to deliver half of the overall current budget at 2.5 V.

To avoid power restrictions for DPS128 operation, you need to consider the layout of the power domains in your applications and DUT board designs.
There are no restrictions when operating a DPS64.

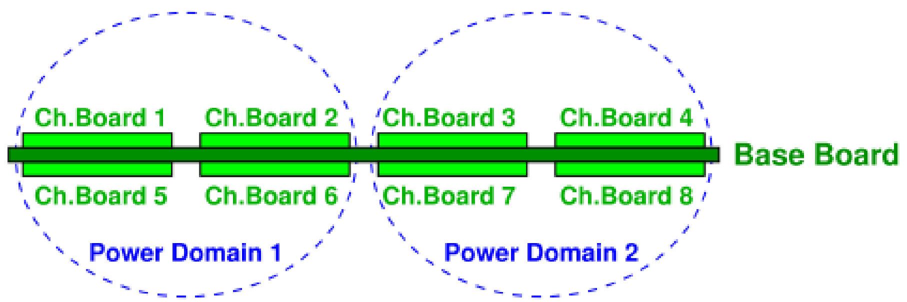


Figure 49: DPS128 power domains

For more details refer to chapter “Power Budget”.

8.3 External Capacitance Range

There are different requirements for the different DPS128 channel types (HC and HV) and the different operation modes (DPS and PMU).

Both channel types require a minimum of 1 nF to 10 nF external capacitance to guarantee stability, mainly targeting improved noise suppression. This requirement is independent of the operation mode.
For more details refer to next chapter “Capacitor Requirements for Improved Noise Suppression”.

In PMU operating mode this 1 nF to 10 nF is sufficient. No further external capacitance required.

In DPS operating mode some additional minimum external load capacitance is required. Overall this will result in improved DPS stability, reduced oscillations and reduced noise.
The requirements differ for HC and HV channels:

- for HC type of channels: 1 µF to 10 µF
- for HV type of channels: 100 nF to 4.7 µF

There is also a defined maximum range in dependency to the channel type and the gang factor:

- for HC type of channels: 470 µF per channel (5 mF if ganged)
- for HV type of channels: 22 µF per channel (47 µF if ganged)

See also the technical specifications for DPS128-HC (TDC: 141967) and DPS128-HV (TDC: 141968).

8.4 Capacitor Requirements for Improved Noise Suppression

For improved noise suppression it is recommended to put 0402 type capacitors between the force and return lines of the DPS channels. As DPS related noise shows a spectral content of 100 MHz+, the capacitance value should be between 1 nF to 10 nF. For VI like applications a maximum of 2.2 nF is recommended.

Use capacitors with the following specification:

Capacitance:	2.2 nF (1 nF to 10 nF)
Package:	0402
Type:	X7R

Take special care to keep the impedance of the decoupling circuit as small as possible. This includes also the parasitic such as trace inductance and via inductance (mounting inductance).

In order to keep the parasitic inductances at the minimum it is recommended to place the capacitor on the opposite side of the pogo block just between the force and return via. This also minimizes any potential noise injection to the board.

If the DUT board layout allows, it is also advantageous to distribute additional capacitors on the power plane. This allows an even better suppression of high-frequency noise. By placing additional capacitors the overall impedance can be lowered.

Take care regarding the vias of these capacitors. It is recommended to have the vias in pads. If the board technology does not allow this, use two vias per pin with wider hole diameter as described in Solution (D) below (Solution (A) is worst, Solution (D) is best).

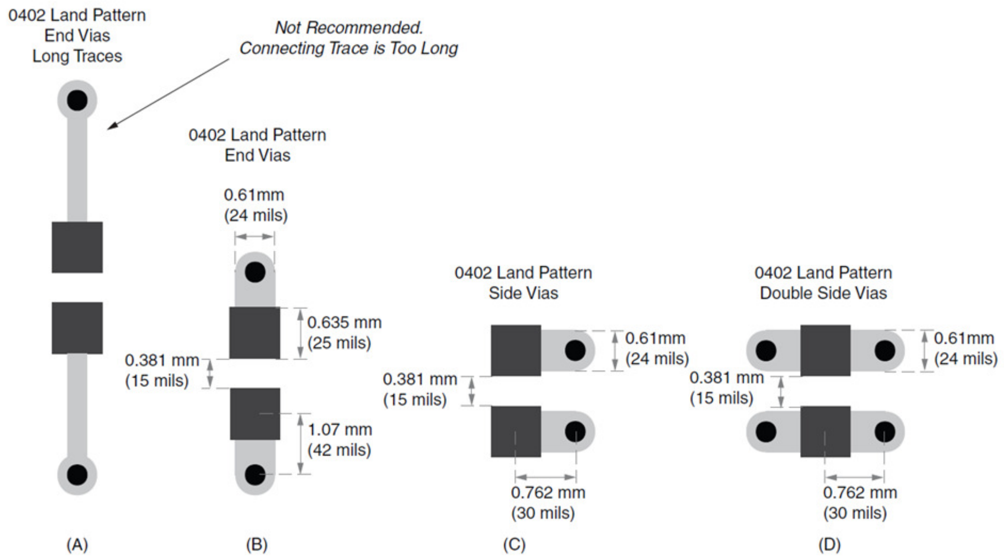


Figure 50: Capacitor mounting examples showing recommended solution (D)

To achieve the best high frequency noise suppression, use special capacitors with an even lower equivalent series inductance like low-ESL LW reverse type capacitors, feed-through (three-terminal) capacitors.

8.5 Current Force Applications

If the DPS128 channel (HC and HV) is used in force current mode with low load impedances (< 2 Ω), it is recommended to add a 1 Ω resistor into the force path before the force-sense connection to avoid possible oscillations.
For more details refer to chapter “Current Force Scenario”.

See also technical specifications for DPS128-HC (TDC: 141967) and DPS128-HV (TDC: 141968).

Appendix

A-1 Model File

There are 2 base types of DPS128 (HC & HV), as well as multiple pogo cables to address any density requirements, as well as compatibility for existing DPS cards like DPS32, MSDPS.

A-1.1 DPS section

In the DPS section of the model file you describe the mapping between DPS type and the channel range (S1 pogo block numbers).

For some detailed syntax description refer to TDC topic 79176.

DPS128 types

```
DCS-DPS128HC | DCS-DPS128HV          #high current or high voltage
```

Example:

```
DPS
20301-20416: type = DCS-DPS128HC    # 32 channels
21101-21816: type = DCS-DPS128HC    #128 channels
```

A-1.2 POGOCABLING section

In the POGOCABLING section of the model file you define the mapping between card cage slots (the physical position of the channel cards) and pogo blocks on the DUT interface. This feature allows flexible routing between the card cage slots and pogo blocks on the DUT interface.

For some detailed syntax description please refer to TDC topic 106944.

DPS128 pogo cables:

<u>E8023PA</u>

- Supports all DPS128 variants
- Standard 128 channel pogo, 8 IO-pogo blocks with 16 channels each

Example:

```
112: pogoblocks="225,425,229,429,303,304,307,308”
```


E8023PC

- Supports E8023CS
- 64 channel pogos, 4 IO-pogo blocks with 16 channels each
- DPS32 compatible, max 2 A per channel
- internal cable ganging of 2 adjacent channels
  - 32 internal channels per IO block (16 channels)
  - 2 logical boards >> needs 1 virtual IO-pogo block address per IO-block
- sequence is defined as (virtual pogo blocks *<u>in italic underline</u>*):
  "a, *<u>a+432</u>*, b, *<u>b+432</u>*, c, *<u>c+432</u>*, d, *<u>d+432</u>*"

Example:

```
112: pogoblocks = "225,657,425,857,229,661,429,861",
type=REDIRECT, subtype=E8023PC,
config="225=225,657; 425=425,857; 229=229,661; 429=429,861"
```

E8023PD

- Supports E8023CSH
- 32 channel pogos, 2 IO-pogo blocks with 16 channels each
- DPS32 compatible, max 2 A per channel
- internal cable ganging of 2 adjacent channels
  - 32 internal channels per IO block (16 channels)
  - 2 logical boards >> needs 1 virtual IO-pogo block address per IO-block
- sequence is defined as (virtual pogo blocks *<u>in italic underline</u>*):
  "a, *<u>a+432</u>*, b, *<u>b+432</u>*"

Example:

```
112: pogoblocks = "225,657,425,857",type=REDIRECT, subtype=E8023PD,
config="225=225,657; 425=425,857"
```

E8023PE

- Support E8023CS
- 64 channel pogos, 8 MSDPS pogo blocks with 8 channels each
- MSDPS compatible, 8 channel mode, max 2 A per channel
- internal cable ganging of 2 adjacent channels
  - 16 internal channels per MSDPS block (8 channels)
  - 1 logical board >> needs 1 virtual IO-pogo block address per MSPDS block
- MSDPS locations are physically used, addresses such as "341, 342" are virtual pogo blocks.
  (Virtual pogo blocks in *<u>italic underline</u>*)

Example:

```
112: pogoblocks = "341,342,343,344,345,346,347,348",
type=REDIRECT, subtype=E8023PE,
config="1=341; 2=342; 3=343; 4=344; 5=345; 6=346; 7=347; 8=348"
```

E8023PF

- Supports E8023CSH
- 16 channel pogos, 2 MSDPS pogo blocks with 8 channels each
- MSDPS compatible, 8 channel mode, max 4 A per channel
- internal cable ganging of 4 adjacent channels
    - 32 internal channels per MSDPS block (8 channels)
    - 2 logical board >> needs 2 virtual IO-pogo block addresses per MSDPS block
- MSDPS locations are physically used, addresses such as “341, 342” are virtual pogo blocks.
  (Virtual pogo blocks in *italic underline*)

Example:

```
112: pogoblocks = "341,342,343,344", type=REDIRECT, subtype=E8023PF,
config="1=341,342; 2=343,344"
```

for first cable

```
113: pogoblocks = "345,346,347,348", type=REDIRECT, subtype=E8023PF,
config="3=345,346; 8=347,348"
```

for second cable

```
114: pogoblocks = "457,458,459,460", type=REDIRECT, subtype=E8023PF,
config="4=457,458; 5=459,460"
```

for third cable

```
115: pogoblocks = "461,462,463,464", type=REDIRECT, subtype=E8023PF,
config="6=461,462; 7=463,464"
```

for fourth cable

E8023PG

- Support E8023CSH
- 32 channel pogos, 4 MSDPS pogo blocks with 8 channels each
- MSDPS compatible, 8 channel mode, max 2 A per channel
- internal cable ganging of 2 adjacent channels
    - 16 internal channels per MSDPS block (8 channels)
    - 1 logical board >> needs 1 virtual IO-pogo block address per MSPDS block
- MSDPS locations are physically used, addresses such as “341, 342” are virtual pogo blocks.
  (Virtual pogo blocks in *italic underline*)

Example:

```
112: pogoblocks = "341,342,343,344", type=REDIRECT, subtype=E8023PG,
config="1=341; 2=342; 3=343; 4=344"
```


A-2 Pogo Cable Channel Mapping

A-2.1 E8023PA

E8023PA Supported by: - E8023CS - E8023CSH - E8023CSW - E8023CSV	- 128 channels on pogo side, 1 A @ 2.5 V - 8 IO pogo blocks, 16 channels per block, DPS32 pogo compatible

Each channel on the DPS side of the cable is routed straight to the pogo block.

Channels 1-16 on channel module 1 are routed to the first pogo block; channels 17-32 on channel module 2 are routed to the second pogo block and so on. (TDC topic: 141739)

Block A	Block B	Block C	Block D
A-1	B-1	C-1	D-1
A-2	B--2	C-2	D-2
A-3	B-3	C-3	D-3
A-4	B-4	C-4	D-4
A-5	B-5	C-5	D-5
A-6	B-6	C-6	D-6
A-7	B-7	C-7	D-7
A-8	B-8	C-8	D-8
A-9	B-9	C-9	D-9
A-10	B-10	C-10	D-10
A-11	B-11	C-11	D-11
A-12	B-12	C-12	D-12
A-13	B-13	C-13	D-13
A-14	B-14	C-14	D-14
A-15	B-15	C-15	D-15
A-16	B-16	C-16	D-16

Block E	Block F	Block G	Block H
E-1	F-1	G-1	H-1
E-2	F-2	G-2	H-2
E-3	F-3	G-3	H-3
E-4	F-4	G-4	H-4
E-5	F-5	G-5	H-5
E-6	F-6	G-6	H-6
E-7	F-7	G-7	H-7
E-8	F-8	G-8	H-8
E-9	F-9	G-9	H-9
E-10	F-10	G-10	H-10
E-11	F-11	G-11	H-11
E-12	F-12	G-12	H-12
E-13	F-13	G-13	H-13
E-14	F-14	G-14	H-14
E-15	F-15	G-15	H-15
E-16	F-16	G-16	H-16

Figure 51: E8023PA channel mapping table

	Assignment			
Pin	a	b	c	d
X-1	GS2	P1	GND	PS1
X-2	PS2	GND	P2	GS1
X-3	GS4	P3	GND	PS3
X-4	PS4	GND	P4	GS3
X-5	GS6	P5	GND	PS5
X-6	PS6	GND	P6	GS5
X-7	GS8	P7	GND	PS7
X-8	PS8	GND	P8	GS7
X-9	GS10	P9	GND	PS9
X-10	PS10	GND	P10	GS9
X-11	GS12	P11	GND	PS11
X-12	PS12	GND	P12	GS11
X-13	GS14	P13	GND	PS13
X-14	PS14	GND	P14	GS13
X-15	GS16	P15	GND	PS15
X-16	PS16	GND	P16	GS15

Figure 52: E8023PA pogo block layout (DPS32 compatible)

A-2.2 E8023PC

E8023PC Supported by: - E8023CS	- 64 channels on pogo side, 2 A @ 2.5 V (cable ganged) - 4 IO pogo blocks with 16 channels per block, DPS32 pogo compatible

Two DPS128 channels are ganged in the pogo block and routed as one channel.

Channels 1-16 on channel module 1 and channels 17-32 on channel module 2 are routed to the first pogo block; channels 33-48 on channel module 3 and channels 49-64 on channel module 4 are routed to the second pogo block and so on. (TDC topic: 141738)

Block A DPS 128	Block A IO block	Block B DPS 128	Block B IO block	Block C DPS 128	Block C IO block	Block D DPS 128	Block D IO block
A-1 A-2	A-1	C-1 C-2	B-1	E-1 E-2	C-1	G-1 G-2	D-1
A-3 A-4	A--2	C-3 C-4	B-2	E-3 E-4	C-2	G-3 G-4	D-2
A-5 A-6	A-3	C-5 C-6	B-3	E-5 E-6	C-3	G-5 G-6	D-3
A-7 A-8	A-4	C-7 C-8	B-4	E-7 E-8	C-4	G-7 G-8	D-4
A-9 A10	A-5	C-9 C-10	B-5	E-9 E-10	C-5	G-9 G-10	D-5
A-11 A-12	A-6	C-11 C-12	B-6	E-11 E-12	C-6	G-11 G-12	D-6
A-13 A-14	A-7	C-13 C-14	B-7	E-13 E-14	C-7	G-13 G-14	D-7
A-15 A-16	A-8	C-15 C-16	B-8	E-15 E-16	C-8	G-15 G-16	D-8
B-1 B-2	A-9	D-1 D-2	B-9	F-1 F-2	C-9	H-1 H-2	D-9
B-3 B-4	A-10	D-3 D-4	B-10	F-3 F-4	C-10	H-3 H-4	D-10
B-5 B-6	A-11	D-5 D-6	B-11	F-5 F-6	C-11	H-5 H-6	D-11
B-7 B-8	A-12	D-7 D-8	B-12	F-7 F-8	C-12	H-7 H-8	D-12
B-9 B-10	A-13	D-9 D-10	B-13	F-9 F-10	C-13	H9 H-10	D-13
B-11 B-12	A-14	D-11 D-12	B-14	F-11 F-12	C-14	H-11 H-12	D-14
B-13 B-14	A-15	D-13 D-14	B-15	F-13 F-14	C-15	H-13 H-14	D-15
B-15 B-16	A-16	D-15 D-16	B-16	F-15 F-16	C-16	H-15 H-16	D-16

Figure 53: E8023PC channel mapping table

	Pin	Assignment a	b	c	d
Channel module A / C / H / F	X-1	GS2 (GS3+GS4)	P1 (P1+P2)	GND	PS1 (PS1+PS2)
	X-2	PS2 (PS3+PS4)	GND	P2 (P3+P4)	GS1 (GS1+GS2)
	X-3	GS4 (GS7+GS8)	P3 (P5+P6)	GND	PS3 (PS5+PS6)
	X-4	PS4 (PS7+PS8)	GND	P4 (P7+P8)	GS3 (GS5+GS6)
	X-5	GS6 (GS11+GS12)	P5 (P9+P10)	GND	PS5 (PS9+PS10)
	X-6	PS6 (PS11+PS12)	GND	P6 (P11+P12)	GS5 (GS9+GS10)
	X-7	GS8 (GS15+GS16)	P7 (P13+P14)	GND	PS7 (PS13+PS14)
	X-8	PS8 (PS15+PS16)	GND	P8 (P15+P16)	GS7 (GS13+GS14)
Channel module B / D / G / E	X-9	GS10 (GS3+GS4)	P9 (P1+P2)	GND	PS9 (PS1+PS2)
	X-10	PS10 (PS3+PS4)	GND	P10 (P3+P4)	GS9 (GS1+GS2)
	X-11	GS12 (GS7+GS8)	P11 (P5+P6)	GND	PS11 (PS5+PS6)
	X-12	PS12 (PS7+PS8)	GND	P12 (P7+P8)	GS11 (GS5+GS6)
	X-13	GS14 (GS11+GS12)	P13 (P9+P10)	GND	PS13 (PS9+PS10)
	X-14	PS14 (PS11+PS12)	GND	P14 (P11+P12)	GS13 (GS9+GS10)
	X-15	GS16 (GS15+GS16)	P15 (P13+P14)	GND	PS15 (PS13+PS14)
	X-16	PS16 (PS15+PS16)	GND	P16 (P15+P16)	GS15 (GS13+GS14)

Figure 54: E8023PC pogo block layout (DPS32 compatible)

A-2.3 E8023PD

E8023PD Supported by: - E8023CSH	- 32 channels on pogo side, 2 A @ 2.5 V (cable ganged) - 2 IO pogo blocks with 16 channels per block, DPS32 pogo compatible

Two DPS64 channels are ganged in the pogo block and routed as one channel.

Channels 1-16 on channel module 1 and channels 17-32 on channel module 2 are routed to the first pogo block; channels 33-48 on channel module 3 and channels 49-64 on channel module 4 are routed to the second pogo block and so on. (TDC topic: 141744)

Block A		Block B	
DPS 64	IO block	DPS 64	IO block
A-1 A-2	A-1	C-1 C-2	B-1
A-3 A-4	A--2	C-3 C-4	B-2
A-5 A-6	A-3	C-5 C-6	B-3
A-7 A-8	A-4	C-7 C-8	B-4
A-9 A10	A-5	C-9 C-10	B-5
A-11 A-12	A-6	C-11 C-12	B-6
A-13 A-14	A-7	C-13 C-14	B-7
A-15 A-16	A-8	C-15 C-16	B-8
B-1 B-2	A-9	D-1 D-2	B-9
B-3 B-4	A-10	D-3 D-4	B-10
B-5 B-6	A-11	D-5 D-6	B-11
B-7 B-8	A-12	D-7 D-8	B-12
B-9 B-10	A-13	D-9 D-10	B-13
B-11 B-12	A-14	D-11 D-12	B-14
B-13 B-14	A-15	D-13 D-14	B-15
B-15 B-16	A-16	D-15 D-16	B-16

Figure 55: E8023PD channel mapping table

		Assignment			
	Pin	a	b	c	d
Channel module A / C	X-1	GS2 (GS3+GS4)	P1 (P1+P2)	GND	PS1 (PS1+PS2)
	X-2	PS2 (PS3+PS4)	GND	P2 (P3+P4)	GS1 (GS1+GS2)
	X-3	GS4 (GS7+GS8)	P3 (P5+P6)	GND	PS3 (PS5+PS6)
	X-4	PS4 (PS7+PS8)	GND	P4 (P7+P8)	GS3 (GS5+GS6)
	X-5	GS6 (GS11+GS12)	P5 (P9+P10)	GND	PS5 (PS9+PS10)
	X-6	PS6 (PS11+PS12)	GND	P6 (P11+P12)	GS5 (GS9+GS10)
	X-7	GS8 (GS15+GS16)	P7 (P13+P14)	GND	PS7 (PS13+PS14)
	X-8	PS8 (PS15+PS16)	GND	P8 (P15+P16)	GS7 (GS13+GS14)
Channel module B / D	X-9	GS10 (GS3+GS4)	P9 (P1+P2)	GND	PS9 (PS1+PS2)
	X-10	PS10 (PS3+PS4)	GND	P10 (P3+P4)	GS9 (GS1+GS2)
	X-11	GS12 (GS7+GS8)	P11 (P5+P6)	GND	PS11 (PS5+PS6)
	X-12	PS12 (PS7+PS8)	GND	P12 (P7+P8)	GS11 (GS5+GS6)
	X-13	GS14 (GS11+GS12)	P13 (P9+P10)	GND	PS13 (PS9+PS10)
	X-14	PS14 (PS11+PS12)	GND	P14 (P11+P12)	GS13 (GS9+GS10)
	X-15	GS16 (GS15+GS16)	P15 (P13+P14)	GND	PS15 (PS13+PS14)
	X-16	PS16 (PS15+PS16)	GND	P16 (P15+P16)	GS15 (GS13+GS14)

Figure 56: E8023PD pogo block layout (DPS32 compatible)

A-2.4 E8023PE

E8023PE	-	64 channels on pogo side, 2 A @ 2.5 V (cable ganged)
Supported by:	-	8 MSDPS pogo blocks with 8 channels per block, MSDPS pogo compatible
- E8023CS	-	MSDPS redirect , GULP style

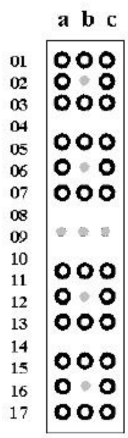
Two DPS128 channels are ganged in the pogo block and routed as one channel.

Channels 1-16 on channel module 1 are routed to the first pogo block; channels 17-32 on channel module 2 are routed to the second pogo block and so on. (TDC topic: 141745)

Block A DPS 128	Block A MS-DPS block	Block B DPS 128	Block B MS-DPS block	Block C DPS 128	Block C MS-DPS block	Block D DPS 128	Block D MS-DPS block
A-1 A-2	A-1	B-1 B-2	B-1	C-1 C-2	C-1	D-1 D-2	D-1
A-3 A-4	A-5	B-3 B-4	B-5	C-3 C-4	C-5	D-3 D-4	D-5
A-5 A-6	A-2	B-5 B-6	B-2	C-5 C-6	C-2	D-5 D-6	D-2
A-7 A-8	A-6	B-7 B-8	B-6	C-7 C-8	C-6	D-7 D-8	D-6
A-9 A10	A-3	B-9 B-10	B-3	C-9 C-10	C-3	D-9 D-10	D-3
A-11 A-12	A-7	B-11 B-12	B-7	C-11 C-12	C-7	D-11 D-12	D-7
A-13 A-14	A-4	B-13 B-14	B-4	C-13 C-14	C-4	D-13 D-14	D-4
A-15 A-16	A-8	B-15 B-16	B-8	C-15 C-16	C-8	D-15 D-16	D-8

Block E DPS 128	Block E MS-DPS block	Block F DPS 128	Block F MS-DPS block	Block G DPS 128	Block G MS-DPS block	Block H DPS 128	Block H MS-DPS block
E-1 E-2	E-1	F-1 F-2	F-1	G-1 G-2	G-1	H-1 H-2	H-1
E-3 E-4	E-5	F-3 F-4	F-5	G-3 G-4	G-5	H-3 H-4	H-5
E-5 E-6	E-2	F-5 F-6	F-2	G-5 G-6	G-2	H-5 H-6	H-2
E-7 E-8	E-6	F-7 F-8	F-6	G-7 G-8	G-6	H-7 H-8	H-6
E-9 E10	E-3	F-9 F-10	F-3	G-9 G-10	G-3	H-9 H-10	H-3
E-11 E-12	E-7	F-11 F-12	F-7	G-11 G-12	G-7	H-11 H-12	H-7
E-13 E-14	E-4	F-13 F-14	F-4	G-13 G-14	G-4	H-13 H-14	H-4
E-15 E-16	E-8	F-15 F-16	F-8	G-15 G-16	G-8	H-15 H-16	H-8

Figure 57:E8023PE channel mapping table



Pin	Assignment a	b	c
X-01	GND	GS5 (GS3+GS4)	GND
X-02	GS1 (GS1+GS2)	NC	PS1 (PS1+PS2)
X-03	P5 (P3+P4)	PS5 (PS3+PS4)	P1 (P1+P2)
X-04			
X-05	GND	GS6 (GS7+GS8)	GND
X-06	GS2 (GS5+GS6)	NC	PS2 (PS5+PS6)
X-07	P6 (P7+P8)	PS6 (PS7+PS8)	P2 (P5+P6)
X-08			
X-09	NC	NC	NC
X-10			
X-11	GND	GS7 (GS11+GS12)	GND
X-12	GS3 (GS9+GS10)	NC	PS3 (PS9+PS10)
X-13	P7 (P11+P12)	PS7 (PS11+PS12)	P3 (P9+P10)
X-14			
X-15	GND	GS8 (GS15+GS16)	GND
X-16	GS4 (GS13+GS14)	NC	PS4 (PS13+PS14)
X-17	P8 (P15+P16)	PS8 (PS15+PS16)	P4 (P13+P14)

Figure 58: E8023PE pogo block layout (MSDSP compatible)

A-2.5 E8023PF

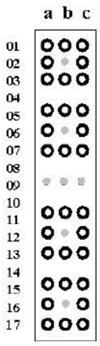
E8023PF	-	16 channels on pogo side, 4 A @ 2.5 V (cable ganged)
Supported by:	-	2 MSDPS pogo blocks with 8 channels per block , MSDPS pogo compatible
- E8023CSH	-	MSDPS redirect , QUL style

Four DPS64 channels are ganged in the pogo block and routed as one channel.

Channels 1-16 on channel module 1 and channels 17-32 on channel module 2 are routed to the first pogo block; channels 33-48 on channel module 3 and channels 49-64 on channel module 4 are routed to the second pogo block. (TDC topic: 141746)

Block A DPS64	Block A MS-DPS block	Block B DPS64	Block B MS-DPS block
A-1	A-1	C-1	B-1
A-2		C-2	
A-3		C-3	
A-4		C-4	
A-5	A--5	C-5	B-5
A-6		C-6	
A-7		C-7	
A-8		C-8	
A-9	A-2	C-9	B-2
A10		C-10	
A-11		C-11	
A-12		C-12	
A-13	A-6	C-13	B-6
A-14		C-14	
A-15		C-15	
A-16		C-16	
B-1	A-3	D-1	B-3
B-2		D-2	
B-3		D-3	
B-4		D-4	
B-5	A-7	D-5	B-7
B-6		D-6	
B-7		D-7	
B-8		D-8	
B-9	A-4	D-9	B-4
B-10		D-10	
B-11		D-11	
B-12		D-12	
B-13	A-8	D-13	B-8
B-14		D-14	
B-15		D-15	
B-16		D-16	

Figure 59:E8023PF channel mapping table



Pin	Assignment a	b	c
X-01	GND	GS5 (GS5+GS6+ GS7+GS8)	GND
X-02	GS1 (GS1+GS2+ GS3+GS4)	NC	PS1 (PS1+PS2+ PS3+PS4)
X-03	P5 (P5+P6+ P7+P8)	PS5 (PS5+PS6+ P7+P8)	P1 (P1+P2+ P3+P4)
X-04			
X-05	GND	GS6 (GS13+GS14+ GS15+GS16)	GND
X-06	GS2 (GS9+GS10+ GS11+GS12)	NC	PS2 (PS9+PS10+ PS11+PS12)
X-07	P6 (P13+P14+ P15+P16)	PS6 (PS13+PS14+ PS15+PS16)	P2 (P9+P10+ P11+P12)
X-08			
X-09	NC	NC	NC
X-10			
X-11	GND	GS7 (GS5+GS6+ GS7+GS8)	GND
X-12	GS3 (GS1+GS2+ GS3+GS4)	NC	PS3 (PS1+PS2+ PS3+PS4)
X-13	P7 (P5+P6+ P7+P8)	PS7 (PS5+PS6+ PS7+PS8)	P3 (P1+P2+ P3+P4)
X-14			
X-15	GND	GS8 (GS13+GS14+ GS15+GS16)	GND
X-16	GS4 (GS9+GS10+ GS11+GS12)	NC	PS4 (PS9+PS10+ PS11+PS12)
X-17	P8 (P13+P14+ P15+P16)	PS8 (PS13+PS14+ PS15+PS16)	P4 (P9+P10+ P11+P12)

Figure 60: E8023PF pogo block layout (MSDSP compatible)

A-2.6 E8023PG

E8023PG	-	32 channels on pogo side, 2 A @ 2.5 V (cable ganged)
Supported by:	-	4 MSDPS pogo blocks with 8 channels per block, MSDPS pogo compatible
- E8023CSH	-	MSDPS redirect , GULP style

Two DPS128 channels are ganged in the pogo block and routed as one channel.

Channels 1-16 on channel module 1 are routed to the first pogo block; channels 17-32 on channel module 2 are routed to the second pogo block and so on. (TDC topic: 140641)

Block A

DPS 128	MS-DPS block
A-1 A-2	A-1
A-3 A-4	A-5
A-5 A-6	A-2
A-7 A-8	A-6
A-9 A10	A-3
A-11 A-12	A-7
A-13 A-14	A-4
A-15 A-16	A-8

Block B

DPS 128	MS-DPS block
B-1 B-2	B-1
B-3 B-4	B-5
B-5 B-6	B-2
B-7 B-8	B-6
B-9 B-10	B-3
B-11 B-12	B-7
B-13 B-14	B-4
B-15 B-16	B-8

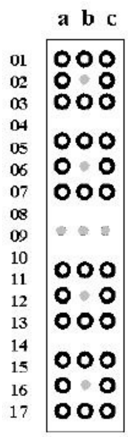
Block C

DPS 128	MS-DPS block
C-1 C-2	C-1
C-3 C-4	C-5
C-5 C-6	C-2
C-7 C-8	C-6
C-9 C-10	C-3
C-11 C-12	C-7
C-13 C-14	C-4
C-15 C-16	C-8

Block D

DPS 128	MS-DPS block
D-1 D-2	D-1
D-3 D-4	D-5
D-5 D-6	D-2
D-7 D-8	D-6
D-9 D-10	D-3
D-11 D-12	D-7
D-13 D-14	D-4
D-15 D-16	D-8

Figure 61:E8023PG channel mapping table



	Assignment		
Pin	a	b	c
X-01	GND	GS5 (GS3+GS4)	GND
X-02	GS1 (GS1+GS2)	NC	PS1 (PS1+PS2)
X-03	P5 (P3+P4)	PS5 (PS3+PS4)	P1 (P1+P2)
X-04			
X-05	GND	GS6 (GS7+GS8)	GND
X-06	GS2 (GS5+GS6)	NC	PS2 (PS5+PS6)
X-07	P6 (P7+P8)	PS6 (PS7+PS8)	P2 (P5+P6)
X-08			
X-09	NC	NC	NC
X-10			
X-11	GND	GS7 (GS11+GS12)	GND
X-12	GS3 (GS9+GS10)	NC	PS3 (PS9+PS10)
X-13	P7 (P11+P12)	PS7 (PS11+PS12)	P3 (P9+P10)
X-14			
X-15	GND	GS8 (GS15+GS16)	GND
X-16	GS4 (GS13+GS14)	NC	PS4 (PS13+PS14)
X-17	P8 (P15+P16)	PS8 (PS15+PS16)	P4 (P13+P14)

Figure 62: E8023PG pogo block layout (MSDSP compatible)

A-3 Model File / Pin Configuration Example for Different Cable Types

This example scenario will focus on DPS128-HC connected with a native E8023PA cable, as well as DPS64-HC connected with DPS32 compatible E8023PD cable and DPS64-HC connected with MSDPS compatible E8023PF cable.

The cables connected to DPS64-HC implement cable ganging and therefore refer to “virtual” pogo locations. In this context “virtual” refers to not being available on the pogo interface. It is highlighted how these virtual locations have to be treated in the model file and the pin configuration, as well as how to setup cable ganging in the pin configuration.

```
MAXIMUM POSSIBLE CONFIGURATION BASED ON THE DETECTED HW:

                 |               |             |          |   Speed     |
    Channel      |      HW       |    Speed    | Vectors  |  Loopback   | misc
-----------------+---------------+-------------+----------+-------------+--------------------
  10101-10816   | PS1600         | 1600 Mbps   |   112M   | 1600 Mbps   | diff, tmu
  20101-20816   | DCS-DPS128HC   | ---------   |    28M   | ---------   |
  30101-30216   | DCS-DPS128HC   | ---------   |    28M   | ---------   |
  34101-34416   | DCS-DPS128HC   | ---------   |    28M   | ---------   |
  73301-73416   | DCS-DPS128HC   | ---------   |    28M   | ---------   |
```

Figure 63: Example Tester Configuration

Based on above configuration report shown during SmarTest start sequence it is not obvious which DPS card type is installed (DPS64 or DPS128). Furthermore it is impossible to gather information about the used pogo cable types (PA, PD or PF).
To extract the DPS type and pogo cable information it is required to have a closer look at the model file.

```
POGOCABLING
  103: pogoblocks="101, 102, 103, 104, 105, 106, 107, 108"
  105: pogoblocks="301,733,302,734", type=REDIRECT,subtype=E8023PD,config="301=301,733;302=302,734"
  106: pogoblocks="341,342,343,344", type=REDIRECT,subtype=E8023PF,config="1=341,342;2=343,344"
  107: pogoblocks="201, 202, 203, 204, 205, 206, 207, 208"

IOCHANNEL
  10101-10816: HW=PS1600, speed= max-rate, smem = 112M, diff, tmu, loopback


DPS
  20101-20816: type = DCS-DPS128HC  # E8023PA

  30101-30216: type = DCS-DPS128HC  # E8023PD
  73301-73416: type = DCS-DPS128HC  # +432

  34101-34416: type = DCS-DPS128HC  # E8023PF
```

Figure 64: Example Model File

DPS relevant data extracted from above model file:

- Channel boards 201 to 208 are connected via E8023PA cable and therefore the board type is DPS128
- Channel boards 301,302 and 733,734 are connected via E8023PD cable and therefore the board type is DPS64. Boards 733 and 734 are virtual, as these hardware channel numbers are not available on the pogo interface
- Channel boards 341 to 344 are connected via E8023PF cable and therefore the board type is DPS64. All boards are virtual board numbers, as they route to MSDPS locations

Note that it is mandatory to specify the virtual channels in the DPS section of the model file, too.

The pin configuration setup is based on the channel pogo block mapping as described in the previous chapter (A-2 Pogo Cable Channel Mapping).
Note that for ganged operation it is required to specify the master as well as the slave channels.

Cable ganged channels can operate in 2 modes

- Ganged: requires specifying the master as well as all necessary slave channels required to fit the maximum current requirement. This includes the virtual channels.
  Master / slave channels are defined by the pogo block mapping of the used cable (refer to chapter A-2)
- Single channel: it is possible to just specify one hardware channel. In this case the cable gang is not "active" and the channel operates in single channel mode

	Name	Pol	C[...	HW	Comment	Tester Channel Site 1
1	PA_20101	POS	0.0	DCS-DPS128HC		20101
2	PA_20301	POS	0.0	DCS-DPS128HC		20301
3	PA_20402_20403	POS	0.0	DCS-DPS128HC	DUT board ganged by2	20402-20403
4	PD_30101	POS	0.0	DCS-DPS128HC	pogo cable ganged by2	30101-30102
5	PD_30302	POS	0.0	DCS-DPS128HC	pogo cable ganged by2	30103-30104
6	PD_30309_30310	POS	0.0	DCS-DPS128HC	pogo cable ganged by2 + DUT board ganged by2	73301-73304
7	PD_30201	POS	0.0	DCS-DPS128HC	pogo cable ganged by2	30201-30202
8	PD_30214	POS	0.0	DCS-DPS128HC	pogo cable ganged by2	73411-73412
9	PD_30215	POS	0.0	DCS-DPS128HC	pogo cable ganged by2 (single channel operation)	73413
10	PD_30216	POS	0.0	DCS-DPS128HC	pogo cable ganged by2 (single channel operation)	73415
11	PF_DPS11	POS	0.0	DCS-DPS128HC	pogo cable ganged by4	34101-34104
12	PF_DPS12	POS	0.0	DCS-DPS128HC	pogo cable ganged by4	34109-34112
13	PF_DPS23	POS	0.0	DCS-DPS128HC	pogo cable ganged by4	34401-34404
14	PF_DPS25	POS	0.0	DCS-DPS128HC	pogo cable ganged by4 (gang by2 operation)	34305-34306
15	PF_DPS28	POS	0.0	DCS-DPS128HC	pogo cable ganged by4 (single channel operation)	34413

Figure 65: Example Pin Configuration

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